

Session 7: From multi locus to whole genome analyses

Molecular Approaches to Clinical Microbiology in Africa 2025

Martin Maiden

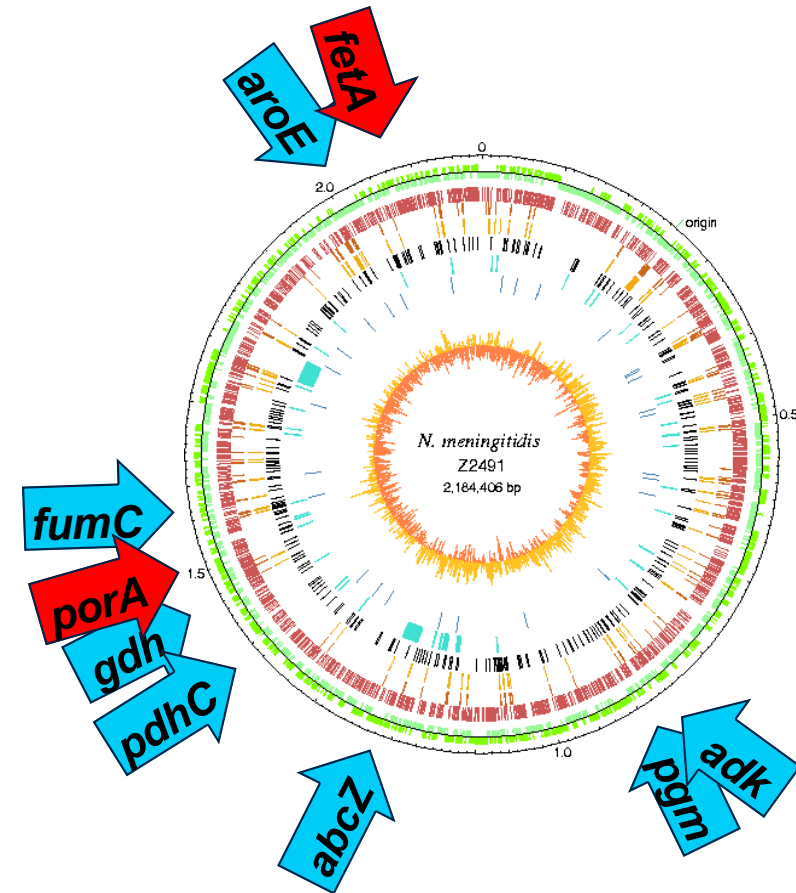
Department of Biology



UNIVERSITY OF
OXFORD

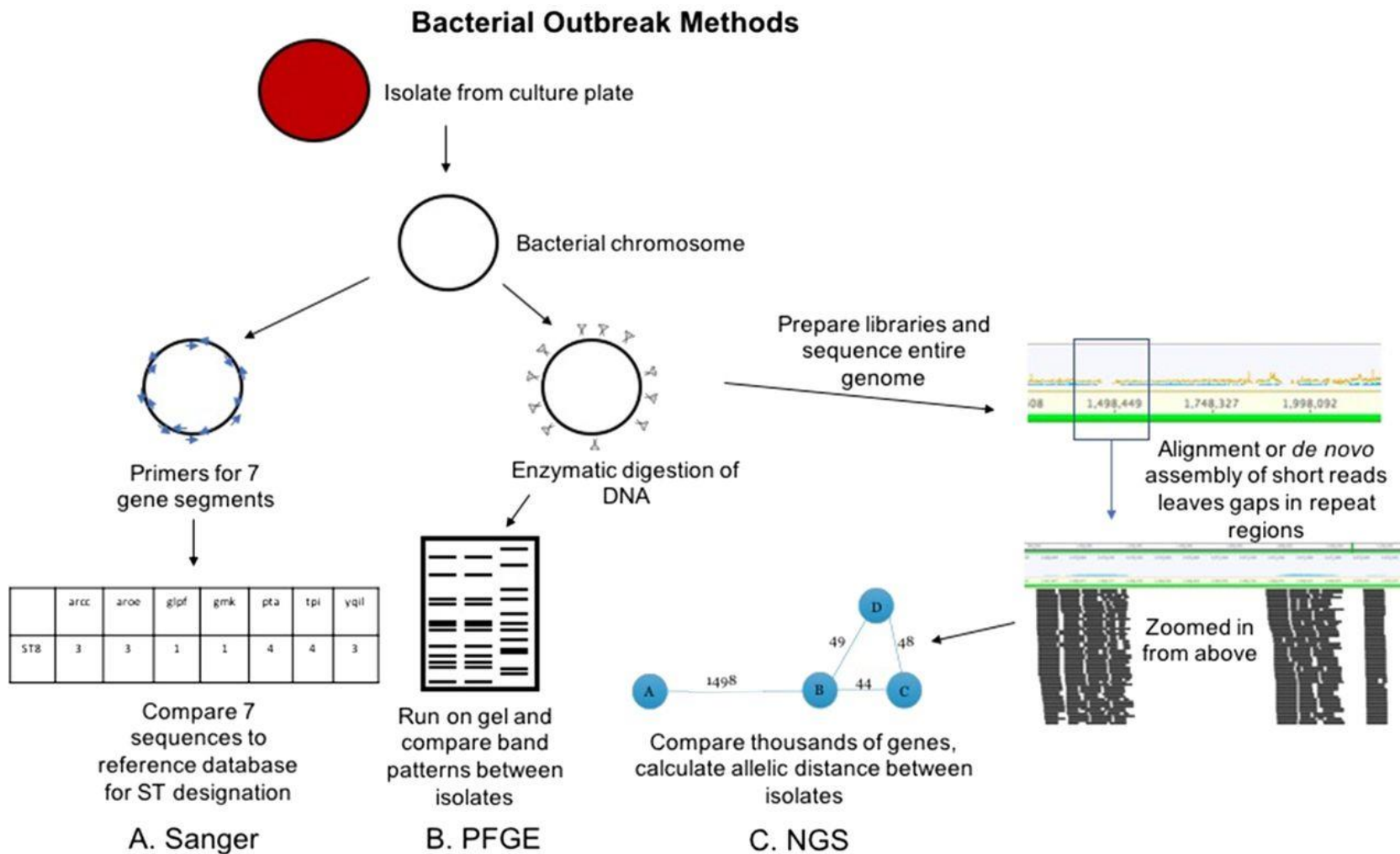
What does whole genome sequence mean?

- ‘Genomic studies’ began with single or multiple gene sequences,
 - e.g. MLST.
- The first whole genome sequences (WGSs) were ‘complete finished genomes’
 - A single contiguous sequence.
- Today, most WGSs are draft assemblies of various quality,
 - aim for a ‘high quality draft’; however, even poor-quality data can be clinically useful.



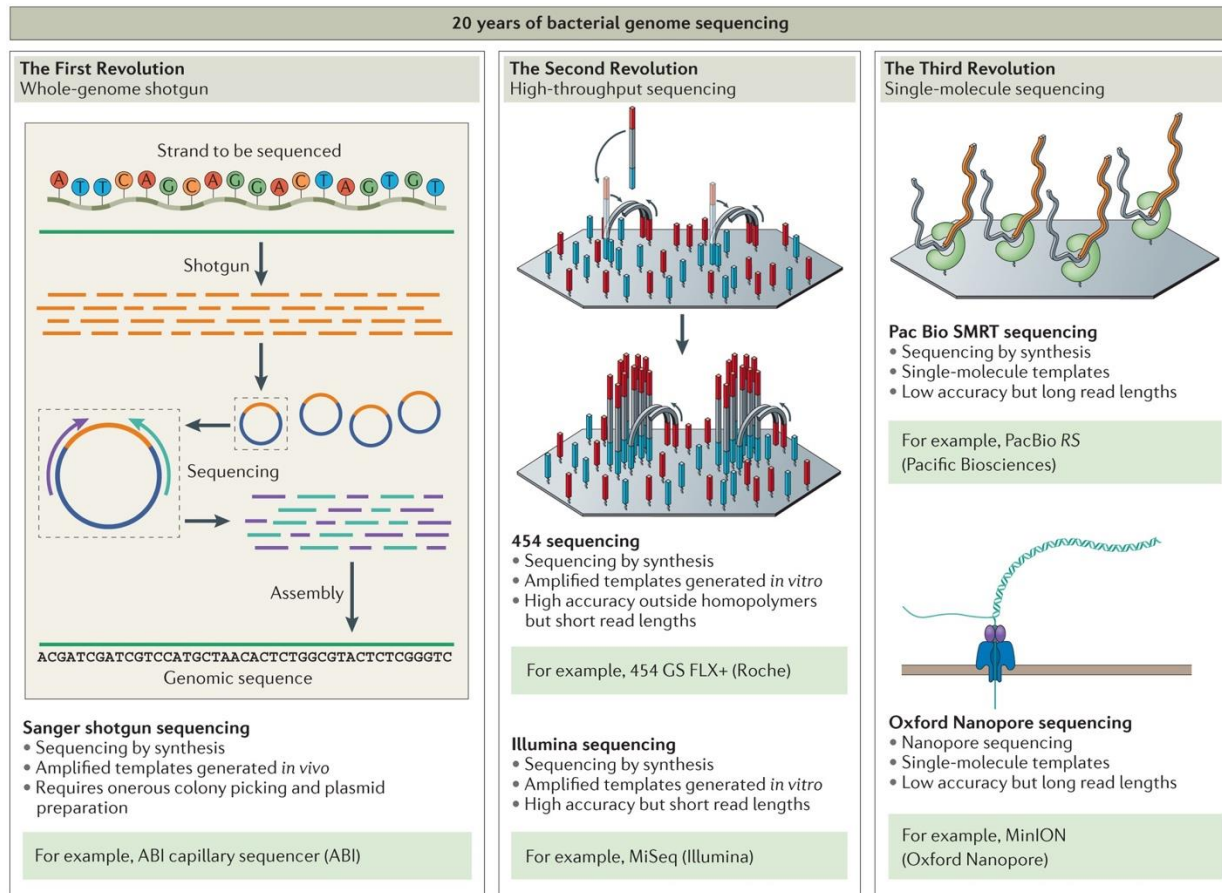
Bratcher, H. B., Corton, C., Jolley, K. A., Parkhill, J. & Maiden, M. C. (2014). A gene-by-gene population genomics platform: *de novo* assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes. *BMC Genomics* **15**, 1138.

Conventional and NGS methods for multi locus bacterial characterisation.



Dougherty, C.E. & Graf, E. (2019). Conventional and NGS methods for bacterial outbreak investigation. *Clin. Lab. Sci.* **32**, 70-77.

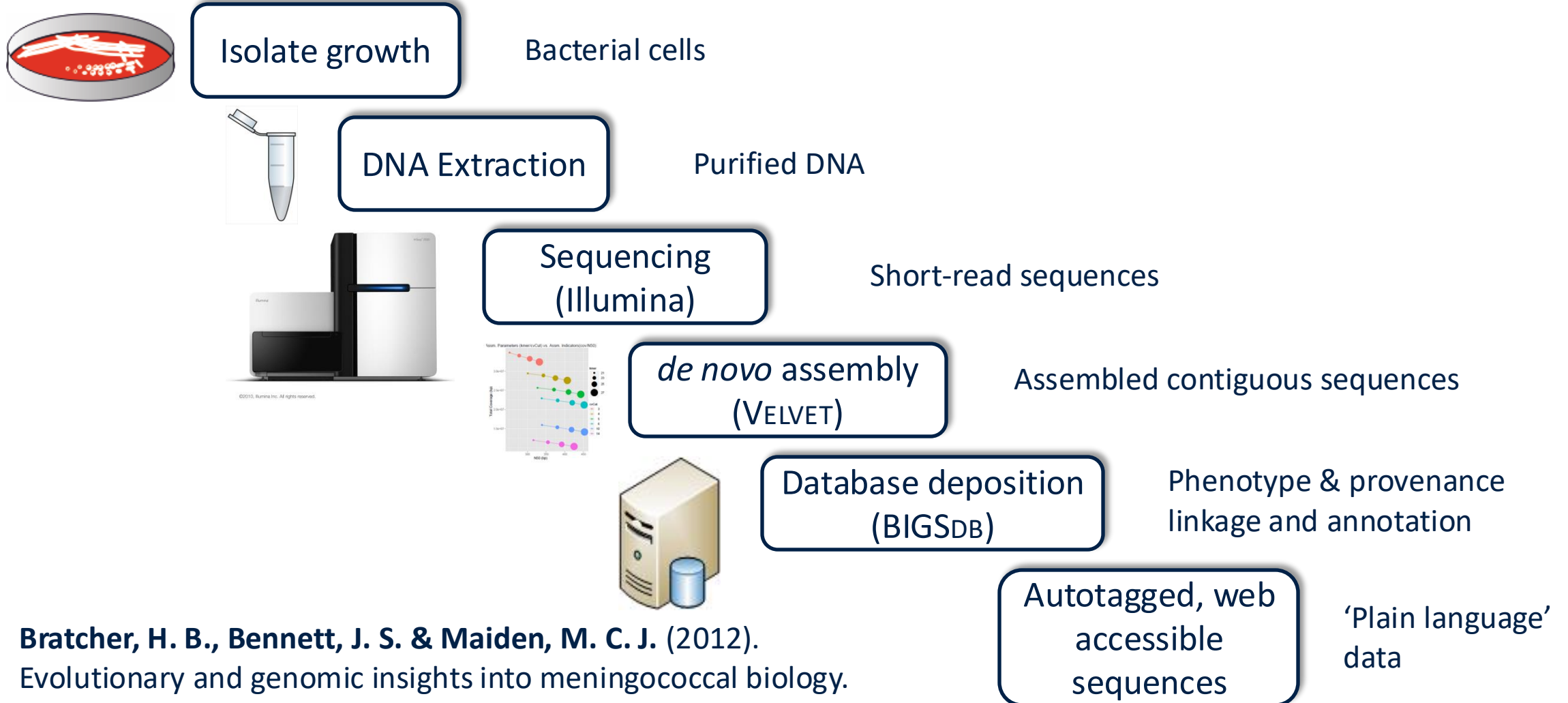
Bacterial Genome Sequencing Approaches



- Three technologies:
 - Sanger; NGS Short-read; & NGS Long-read.
- Computation essential to interpret, these data,
 - assembly software.
- Genomes can be:
 - Complete, closed;
 - Draft – of variable quality ‘high-quality draft’

Loman, N. J. & Pallen, M. J. (2015). Twenty years of bacterial genome sequencing. *Nat Rev Microbiol.* **13**, 787-794.

Whole genome sequence pipeline



Bratcher, H. B., Bennett, J. S. & Maiden, M. C. J. (2012).
Evolutionary and genomic insights into meningococcal biology.
Future Microbiology 7, 873-885.

Approaches to Genome Assembly

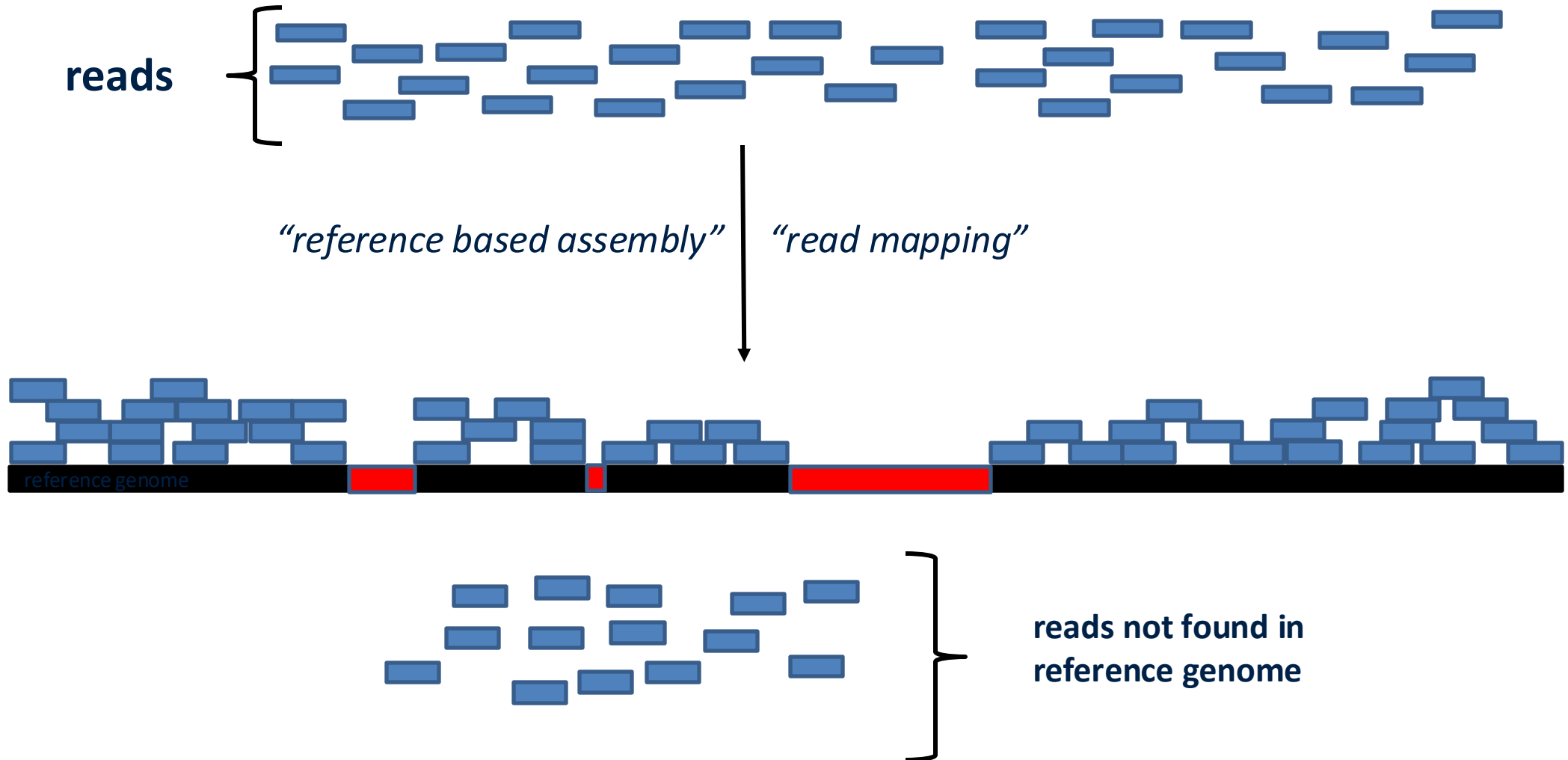
Comparative, reference-based assembly:

- using a reference genome from the same organism or a closely related sample is used to guide the assembly process by aligning the reads;
- primary use is for resequencing applications.

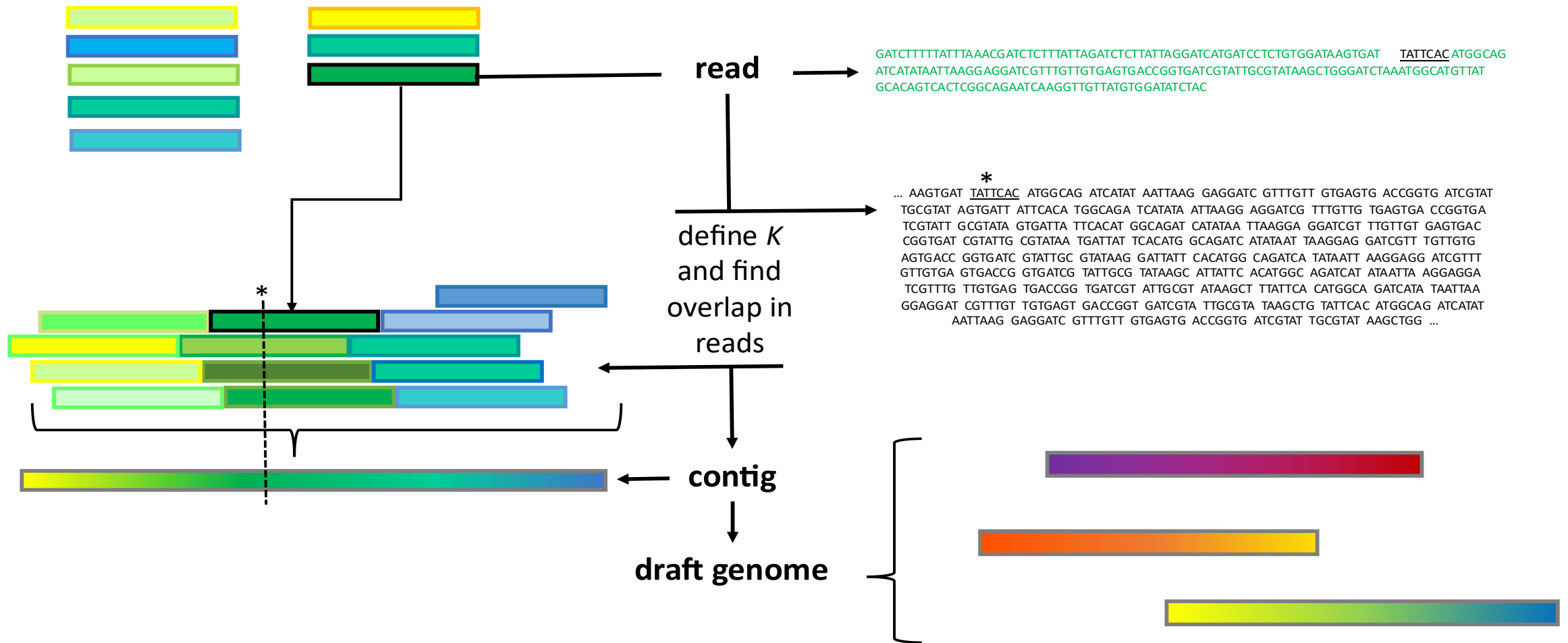
***de novo* assembly:**

- assembly in the strictest sense, no map or guidance is used for assembling the genome;
- used to assemble genomes that have not been previously sequenced.

Read mapping, reference-based assembly



De novo assembly of a draft genome



Assembly gaps

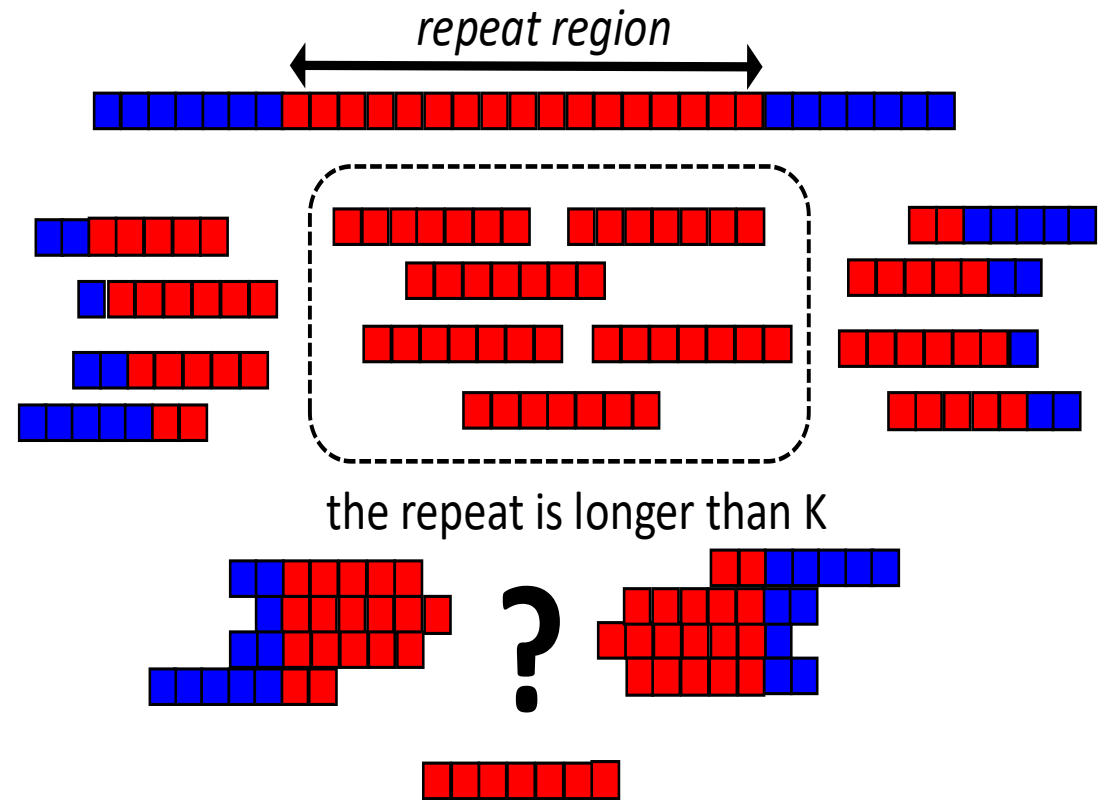
Biased base content:

- AT and GC rich regions are under-represented;
- other chemistry quirks.

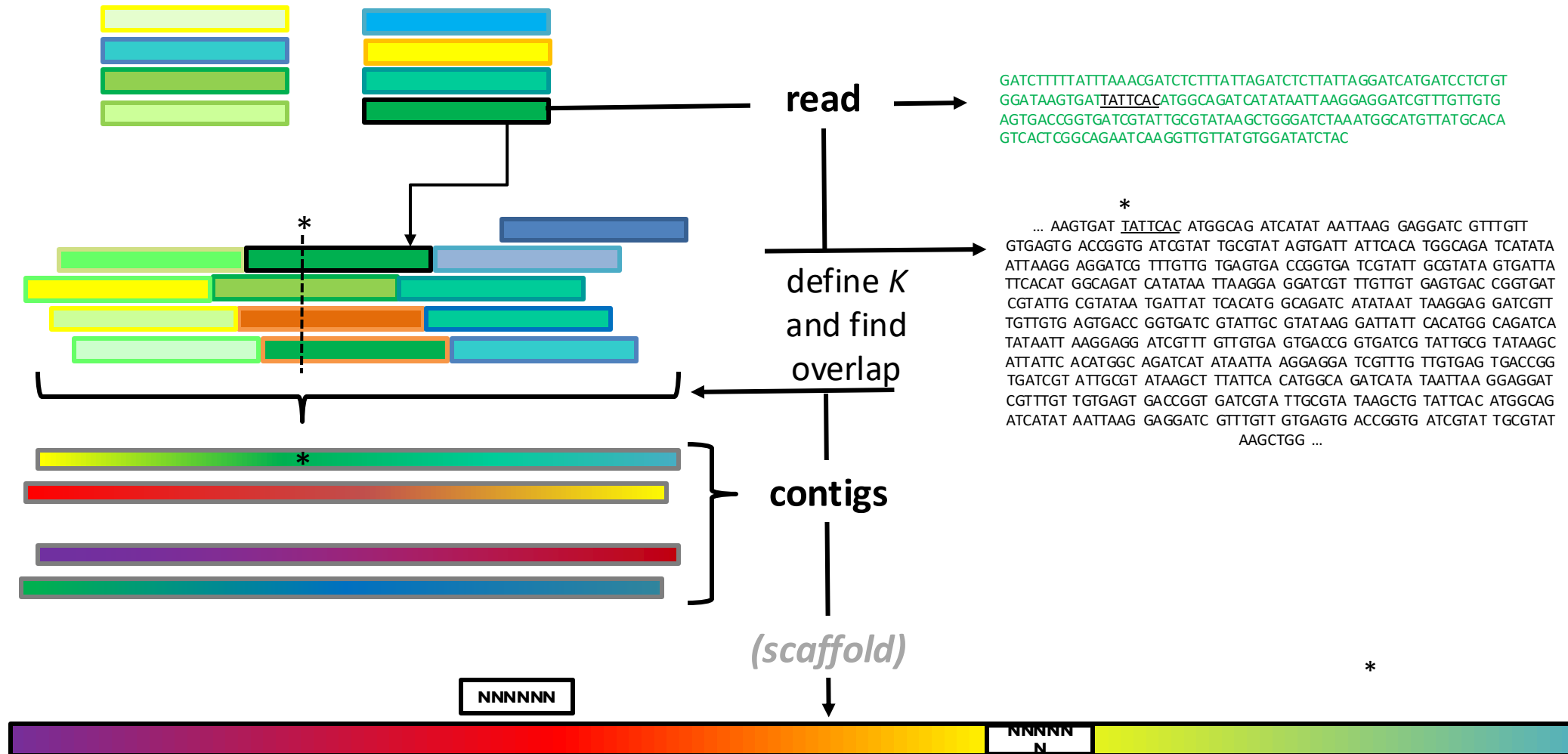
Sequence compression due to repetitive sequences.

➤ Increased depth needed:

- ✓ sequencing errors,
- ✓ polymorphic sites.

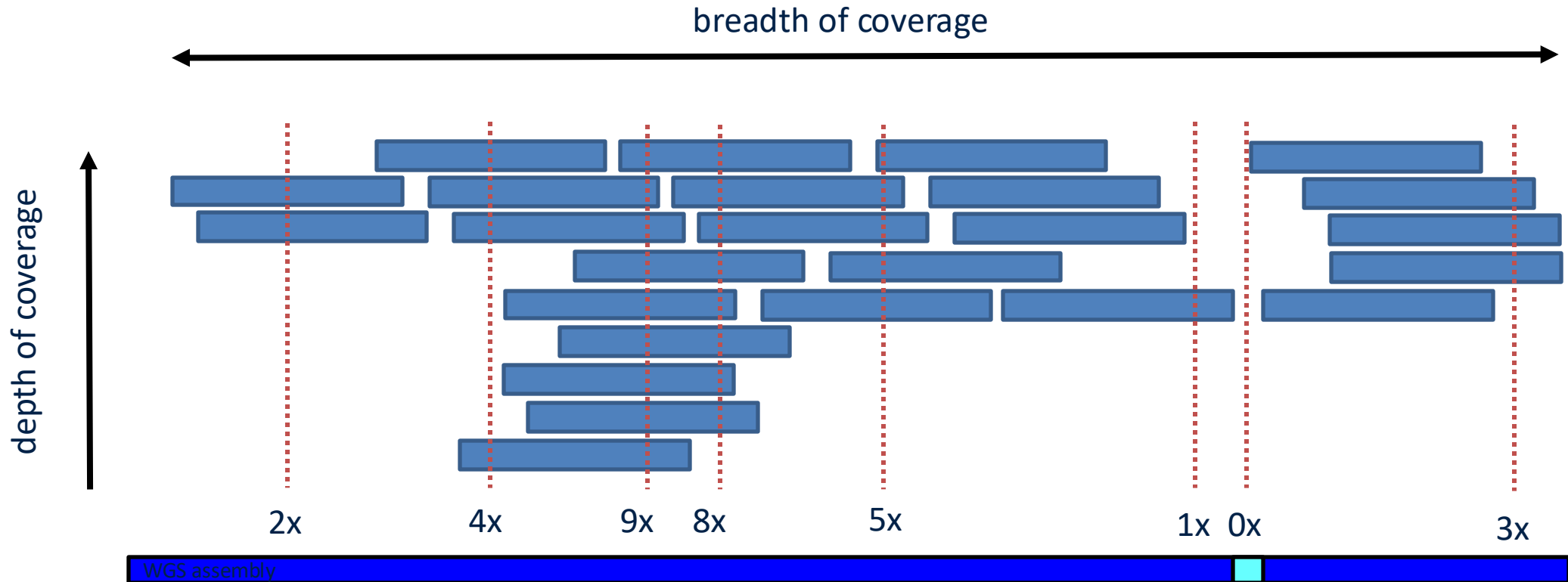


Scaffolding of a draft genome



groups contigs in order but this is a hypothesis that the scaffold is correct

Uniformity of genome coverage



COVERAGE: fraction of the genome sequenced by at least one read

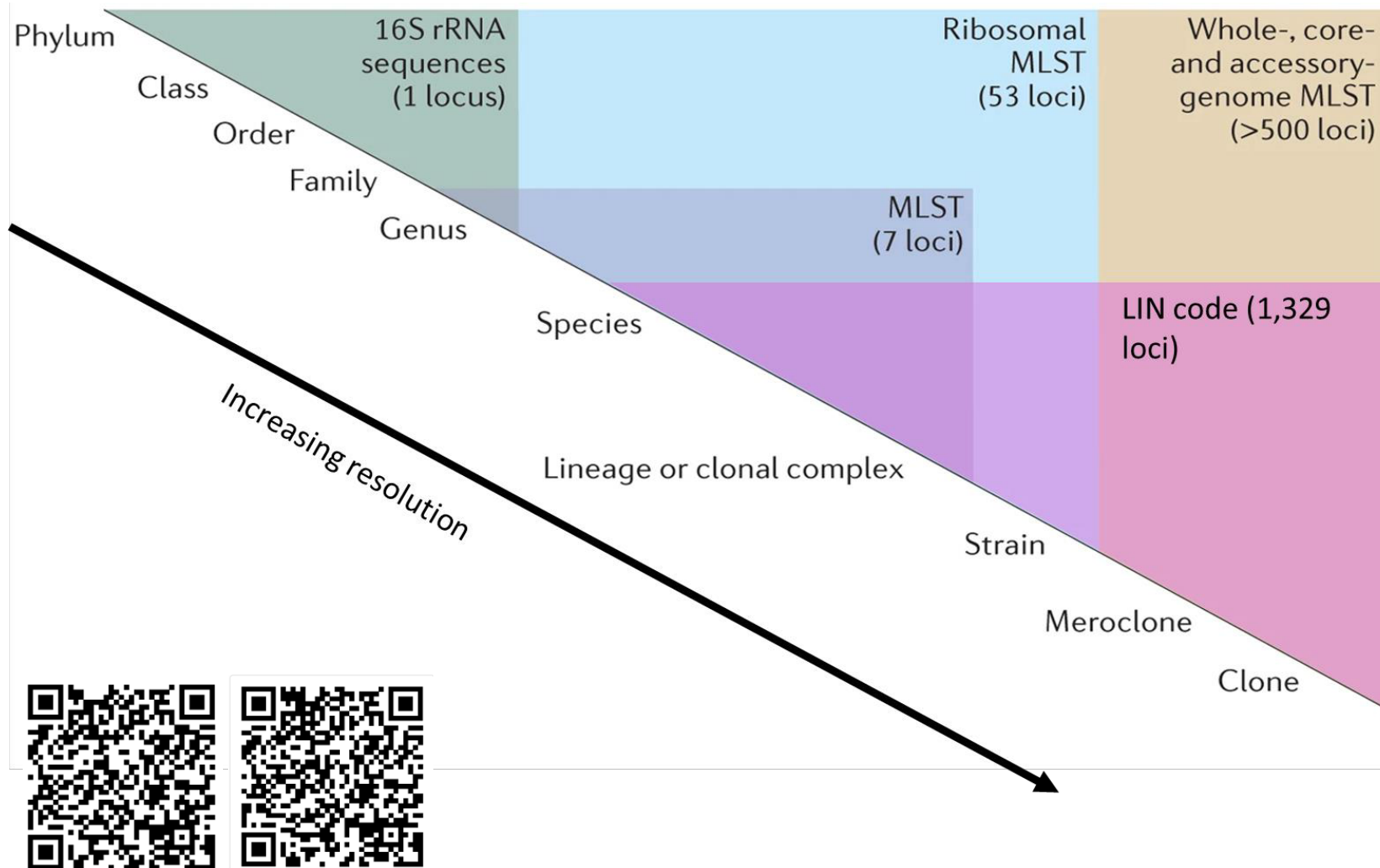
DEPTH: average number of reads that cover any given region

UNIFORMITY: measures the evenness of the coverage depth across the genome

Some quality metrics

Metric	Definition
Number of contigs	Total number of contigs in an assembly
Total length	Combined length of all contigs
Maximum length	Largest contig
Mean length	Average contig size
Coverage	Fraction of the genome sequenced by at least one read
Uniformity	Evenness of the coverage depth across the genome
Depth	Average number of reads that cover any given region
N50	Length of the shortest contig for which longer and equal length contigs cover at least 50 % of the assembly
L50	Number of sequence contigs that are longer than, or equal to, the N50 length and therefore include half the bases of the assembly
N90	Length of the shortest contig for which longer and equal length contigs cover at least 90 % of the assembly

Phenotype, genotype, nomenclature



Maiden, M. C. J., Jansen van Rensburg, M. J., Bray, J. E., Earle, S. G., Ford, S. A., Jolley, K. A. & McCarthy, N. D. (2013). MLST revisited: the gene-by-gene approach to bacterial genomics. *Nature Reviews Microbiology* **11**, 728-736.

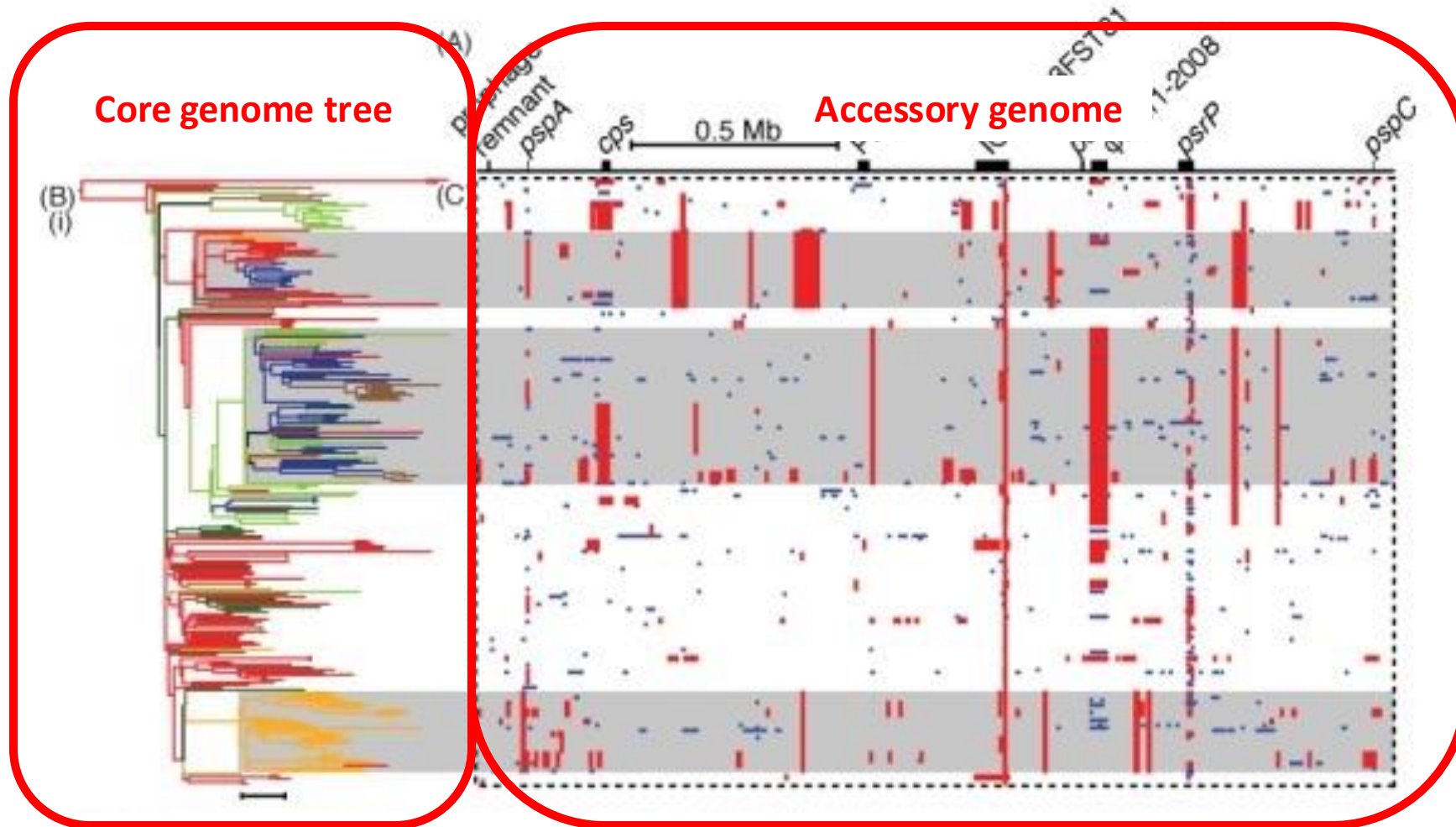
Hennart, M., Guglielmini, J., Bridel, S., Maiden, M. C. J., Jolley, K. A., Criscuolo, A. & Brisse, S. (2022). A Dual Barcoding Approach to Bacterial Strain Nomenclature: Genomic Taxonomy of *Klebsiella pneumoniae* Strains. *Mol Biol Evol.* **39**(7), msac135.

Dealing with partial HGT in WGS

Horizontal gene transfer (HGT, recombination) violates the assumptions of clonal evolution and can be accounted for in models of bacterial evolution by:

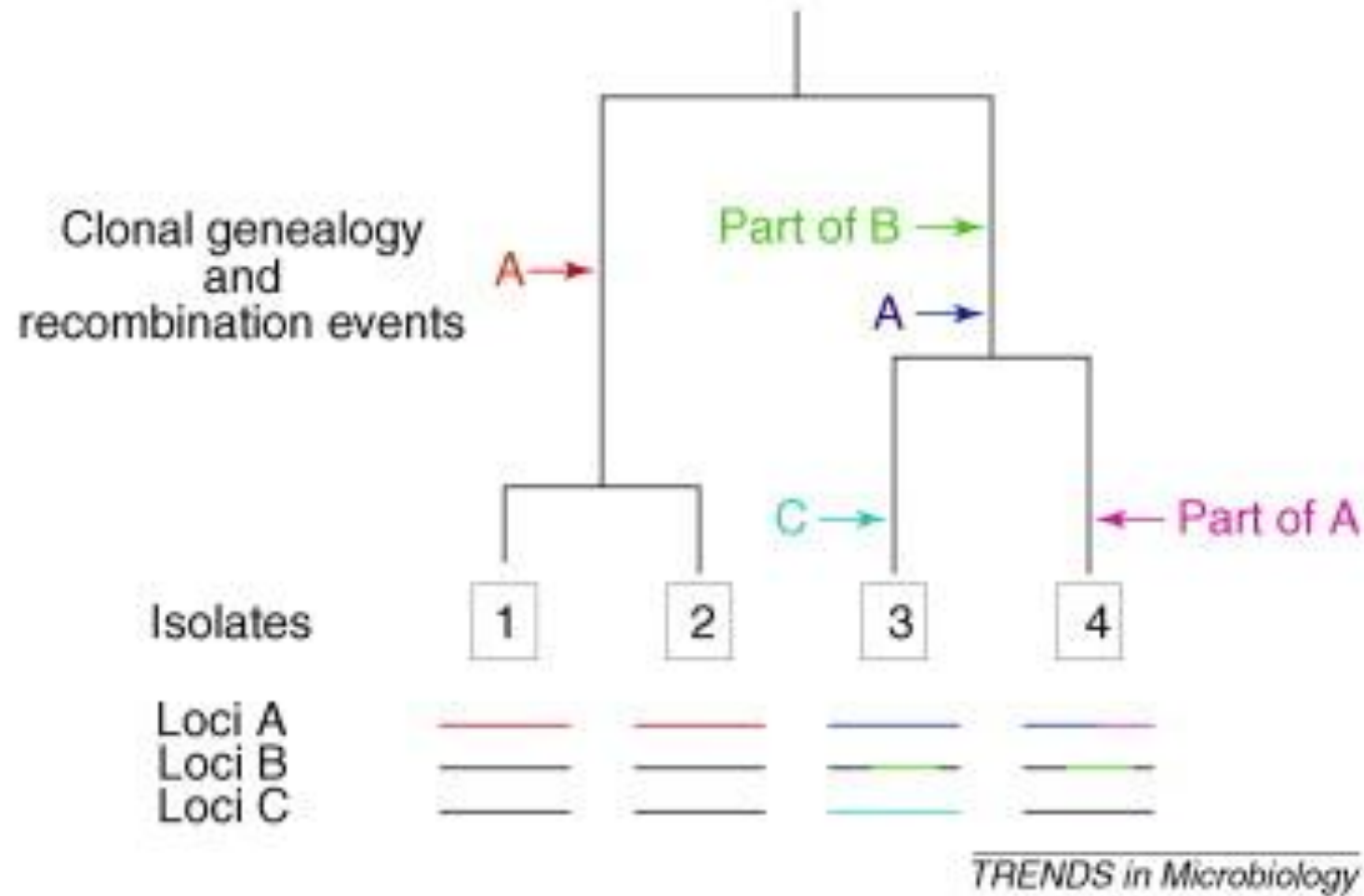
- Ignoring it, which only works if HGT is uncommon;
- Using loci and alleles as the unit of analysis (*e.g.* gene-by-gene MLST approaches);
- Identifying likely HGT and removing it (*e.g.* GUBBINS);
- Estimating relative contributions of recombination and mutation in evolutionary models (*e.g.* ClonalFrame ML).

Gubbins: an analysis of PMEN1 clone



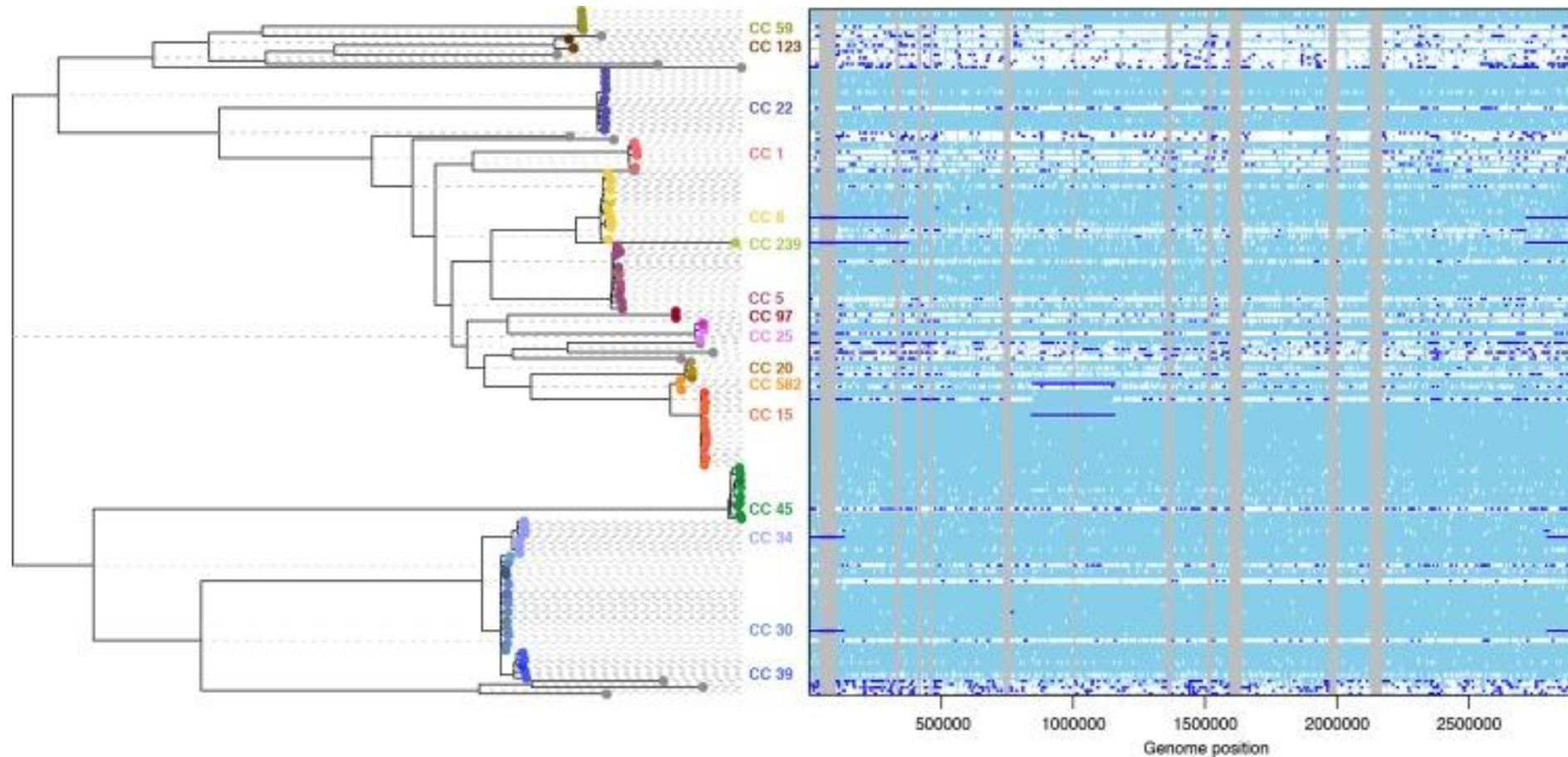
Croucher, N. J., Page, A. J., Connor, T. R., Delaney, A. J., Keane, J. A., Bentley, S. D., Parkhill, J. & Harris, S. R. (2015). Rapid phylogenetic analysis of large samples of recombinant bacterial whole genome sequences using Gubbins. *Nucleic acids research* 43, e15.

The ClonalFrame Model



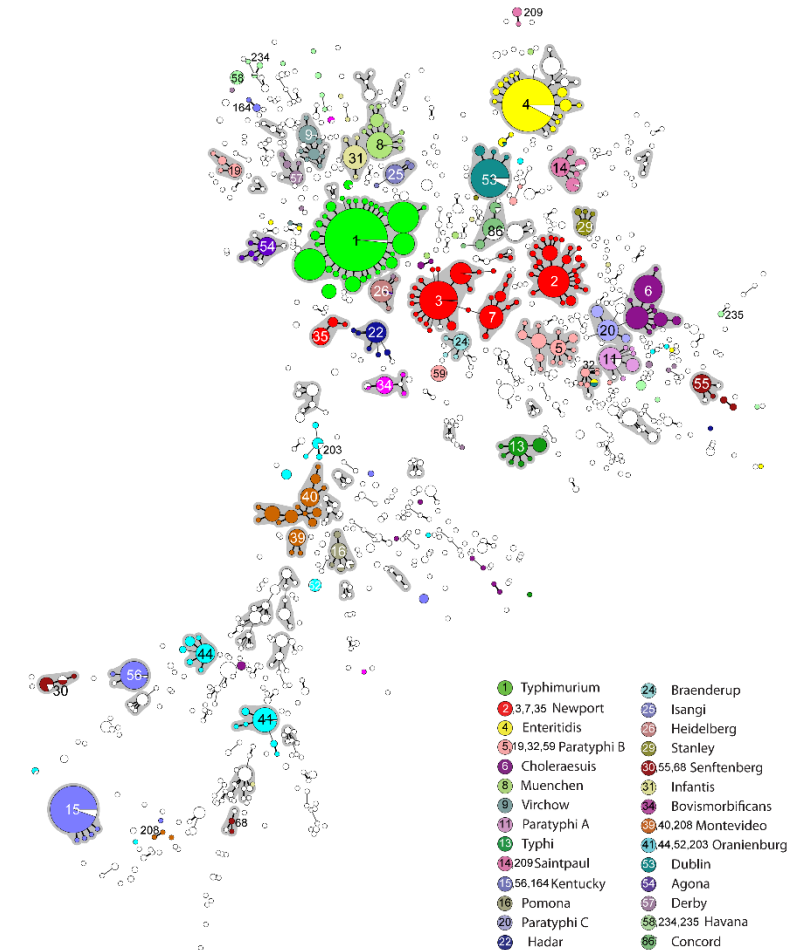
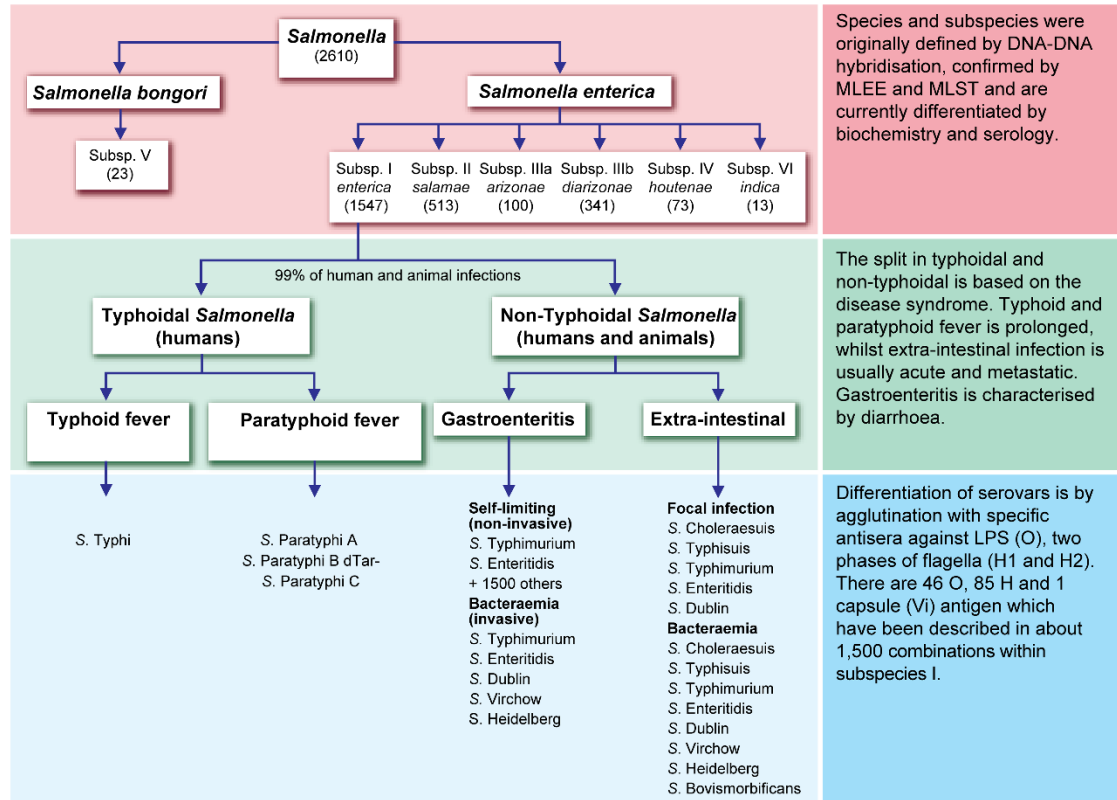
Didelot, X. & Maiden, M. C. (2010). Impact of recombination on bacterial evolution. *Trends in Microbiology* 18, 315-322.

ClonalFrameML: recombination in *S. aureus*



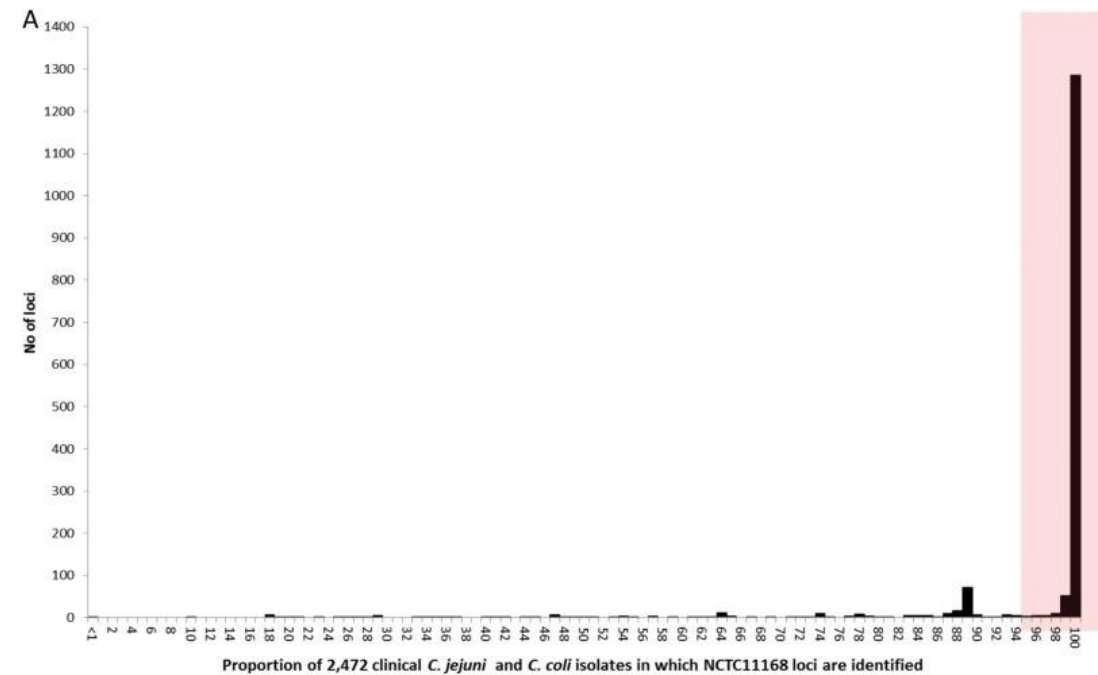
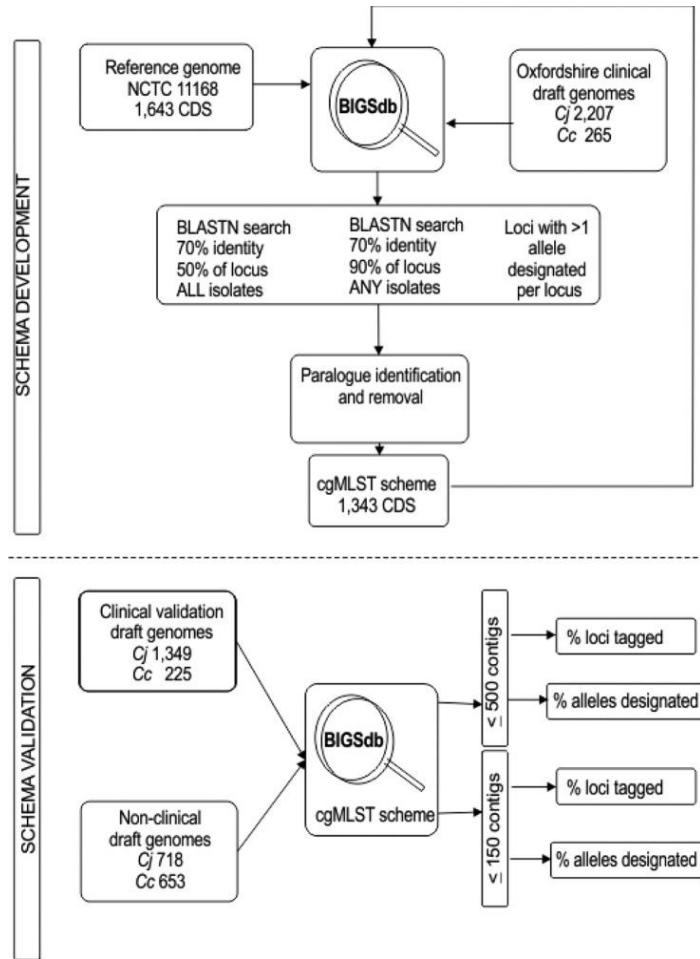
Didelot, X. & Wilson, D. J. (2015). ClonalFrameML: Efficient Inference of Recombination in Whole Bacterial Genomes. *PLoS Computational Biology* 11, e1004041.

Clustering based on alleles: *Salmonella*



Achtman, M., Wain, J., Weill, F. X., Nair, S., Zhou, Z. M., Sangal, V., Krauland, M. G., Hale, J. L., Harbottle, H., Uesbeck, A., Dougan, G., Harrison, L. H., Brisse, S. & SEMS Group (2012). Multilocus Sequence Typing as a Replacement for Serotyping in *Salmonella enterica*. *Plos Pathogens*. 8, e1002776.

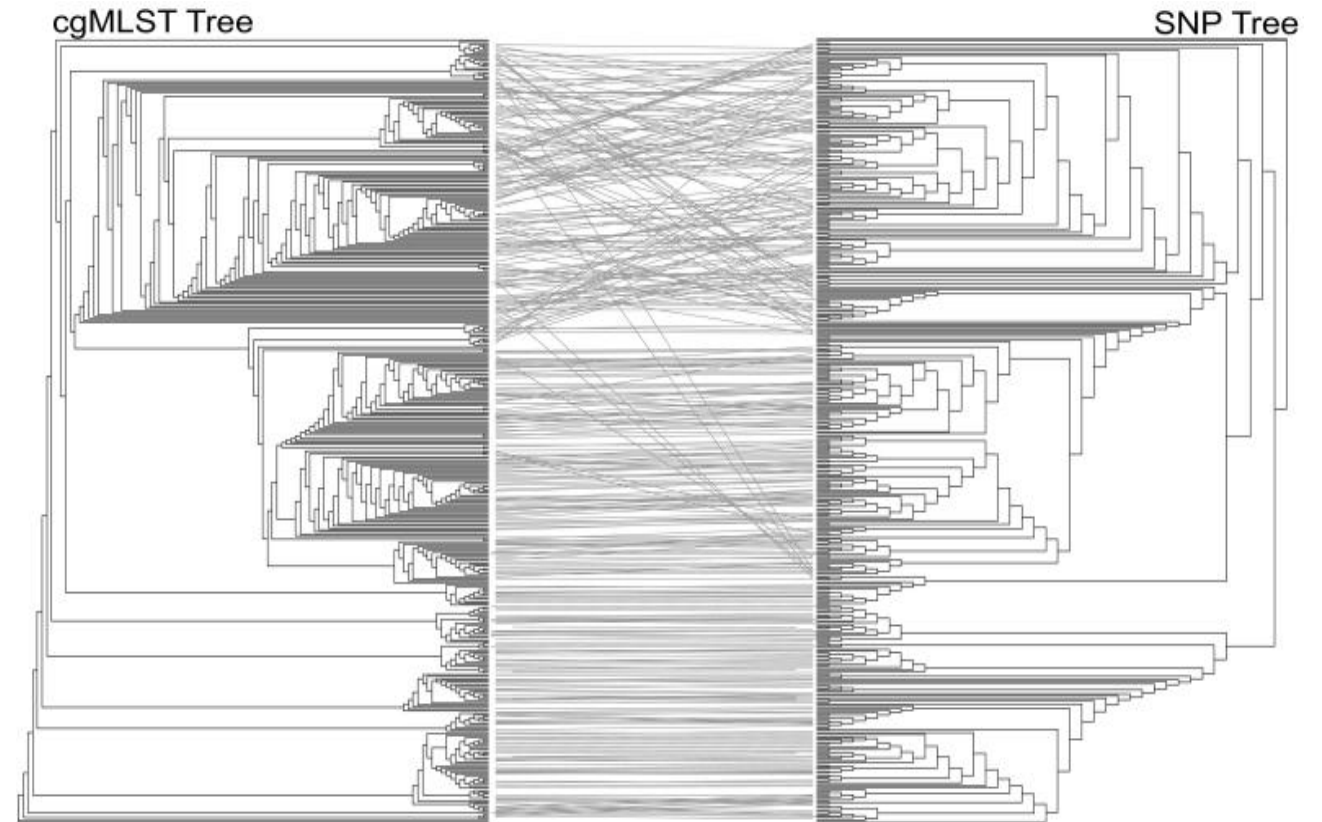
Defining a core genome



Cody, A. J., Bray, J. E., Jolley, K. A., McCarthy, N. D. & Maiden, M. C. J. (2017). Core Genome Multilocus Sequence Typing Scheme for Stable, Comparative Analyses of *Campylobacter jejuni* and *C. coli* Human Disease Isolates. *J Clin Microbiol* 55, 2086-2097.

cgMLST vs 'core SNPs'

- Essentially based on the same data.
- cgMLST can be used to recover 'core SNPs'.
- cgMLST provides a stable nomenclature that is reference free.



Pearce, M. E., Alikhan, N. F., Dallman, T. J., Zhou, Z., Grant, K. & Maiden, M. C. J. (2018). Comparative analysis of core genome MLST and SNP typing within a European *Salmonella* serovar Enteritidis outbreak. *International Journal of Food Microbiology* 274, 1-11.

Gene-by-gene approaches in the analysis of meningococcal genomes

Search or browse database



Enter search criteria or leave blank to browse all records. Modify form parameters to filter or enter a list of values.

Isolate provenance/primary metadata fields

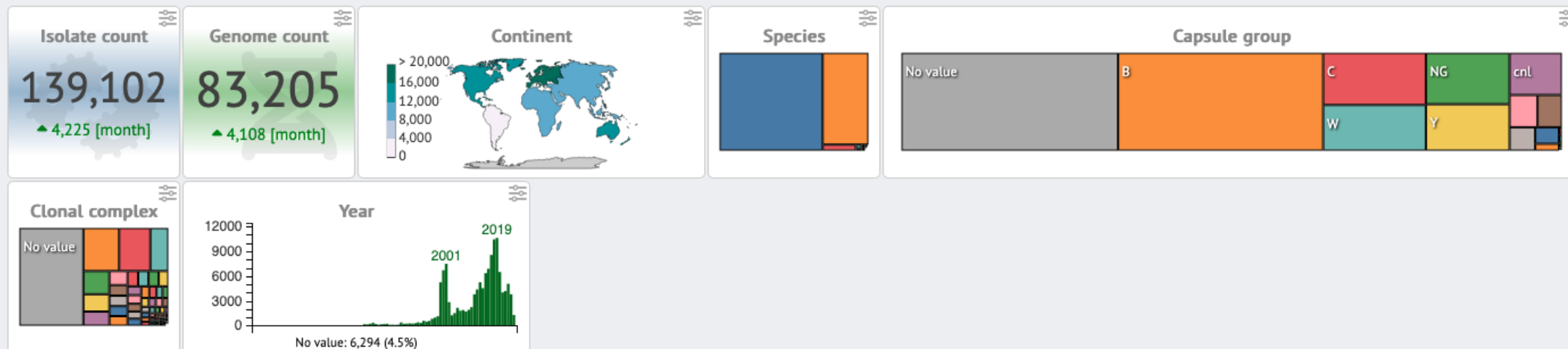
species = e.g. *Neisseria meningitidis* +

Display/sort options

Order by: id ascending

Display: 25 records per page

Action

[RESET](#)
[SEARCH](#)


139,102 records returned (1 - 25 displayed). Click the hyperlinks for detailed information.

Global Genome Library cgMLST



cc10217

cc9316

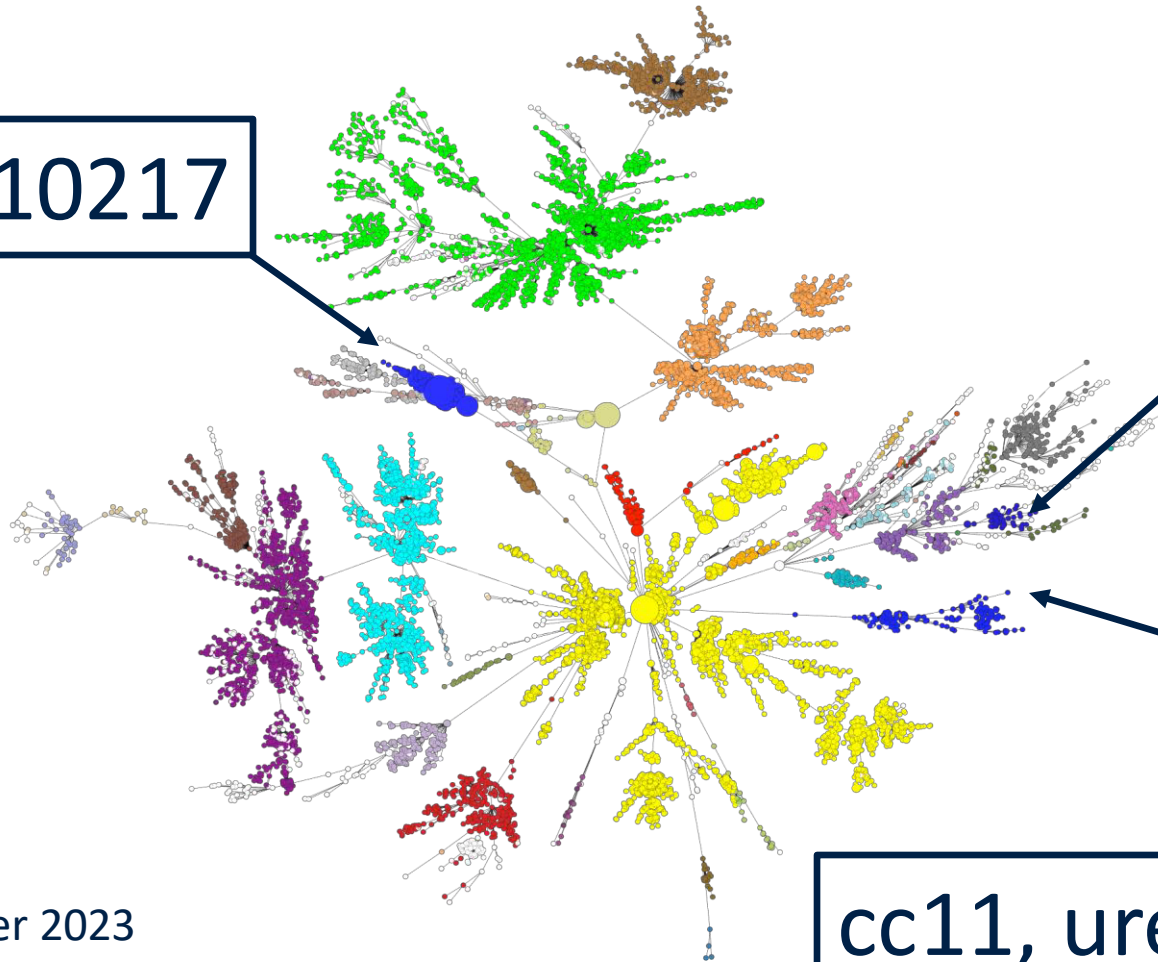
cc4821

cc11, urethritis clone

14,428 Isolates

cgMLST v2.0

Grapetree, 25th September 2023



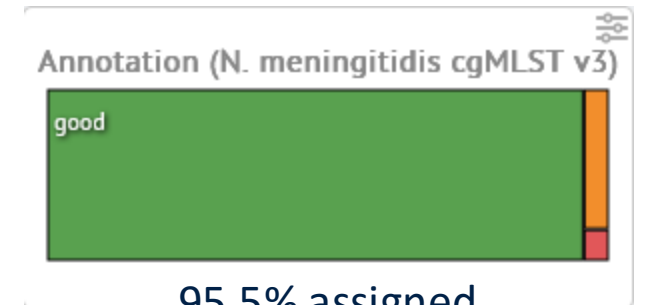
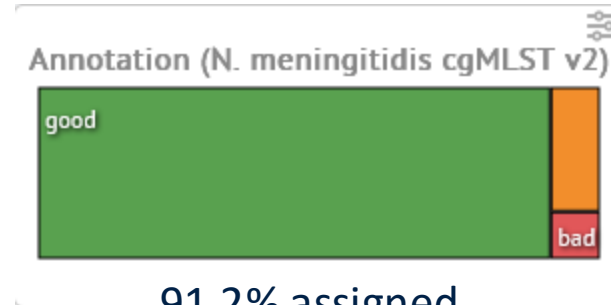
Meningococcal cgMLST schemes

- v1 developed in 2014, 1605 loci,
 - up to 50 missing loci allowed for a cgMLST.
- v2 finalized in 2022 (implemented on PubMLST): 1422 loci,
 - up to 25 missing loci allowed for a cgMLST.
- Problems with v2:
 - inconsistent start sites (especially among different species);
 - multiple consecutive start sites, e.g. ATGATGATG that are not completely conserved;
 - inconsistent end sites where a stop codon is present near the end of a sequence.
- v3 developed by:
 - curating loci to ensure consistent start sites;
 - removing loci that resulted in 'double hits';
 - removing loci that were missing in more than 2% of genomes.
- V3 comprised 1329 loci,
 - up to 25 missing loci allowed for a cgMLST.
- For the purposes of stable typing, the number of loci is less important than their reliability.
- In practise, the differences in the schemes is marginal in terms of resolution.

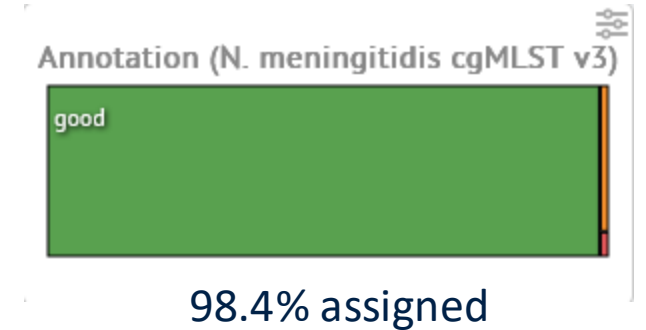
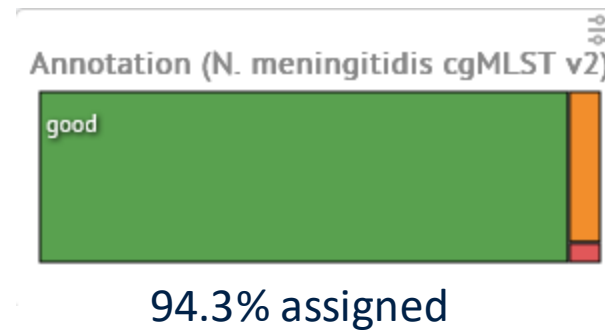
Bratcher, H. B., Corton, C., Jolley, K. A., Parkhill, J. & Maiden, M. C. (2014). A gene-by-gene population genomics platform: *de novo* assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes. *BMC Genomics* **15**, 1138.

Relative performance of cgMLST schemes v2 and v3

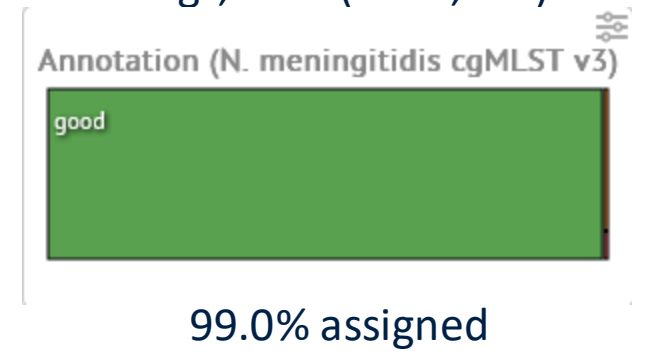
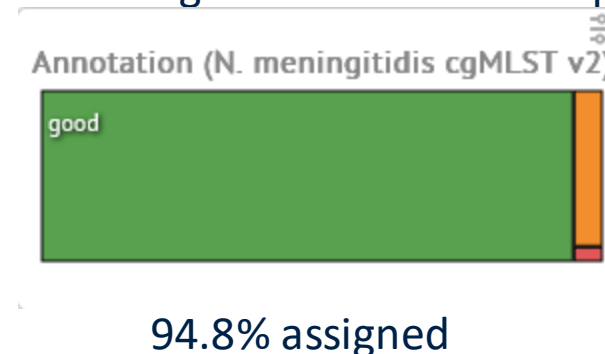
All Nm genomes >2Mbp, 0 Ns (n=41,467)



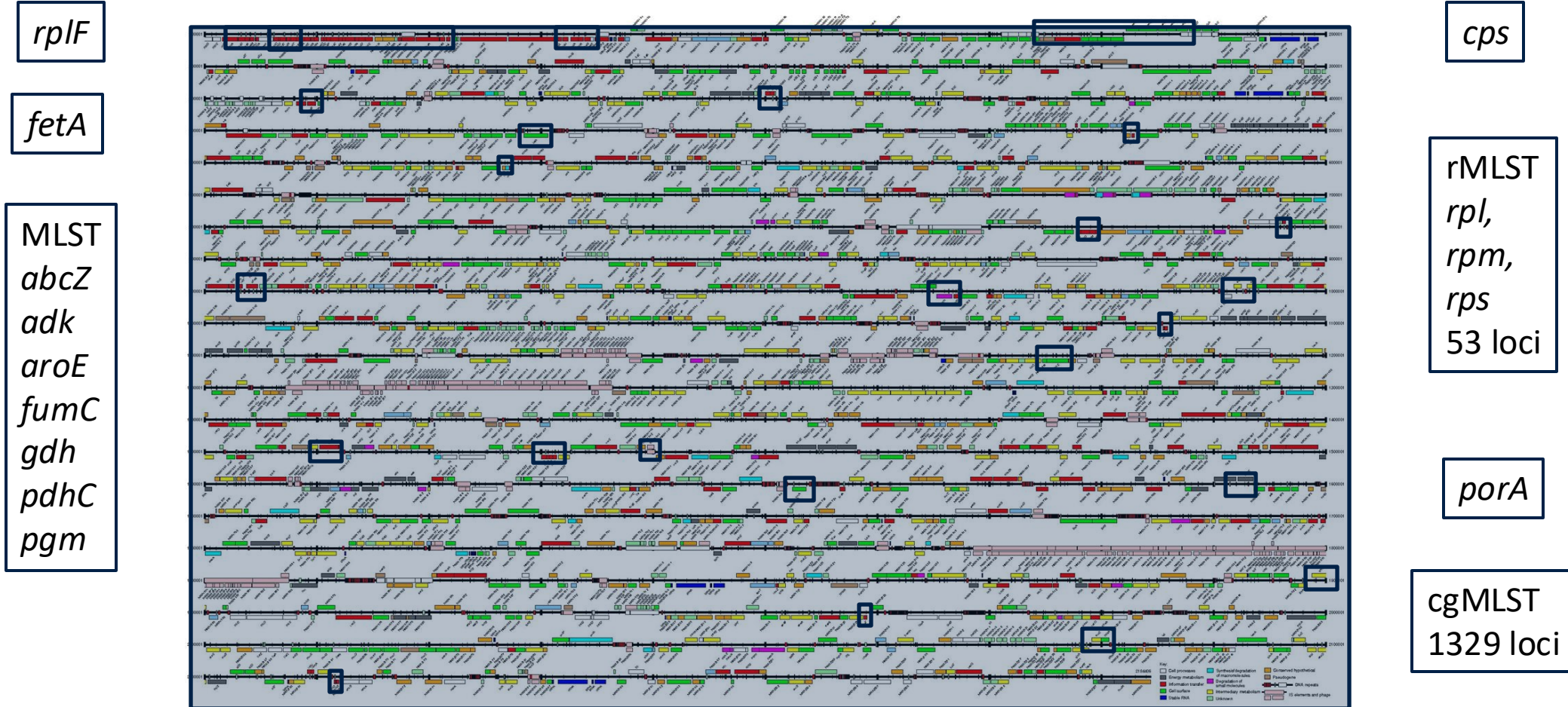
Nm genomes with >2 Mbp, <= 300 contigs, 0 Ns (n=36,409)



Nm genomes with >2 Mbp, <= 200 contigs, 0 Ns (n=29,078)



Summary: typing targets in the meningococcal genome



Parkhill, J., Achtman, M., *et al.* Spratt, B. G. & Barrell, B. G. (2000). Complete DNA sequence of a serogroup A strain of *Neisseria meningitidis* Z2491. *Nature* **404**, 502-506.

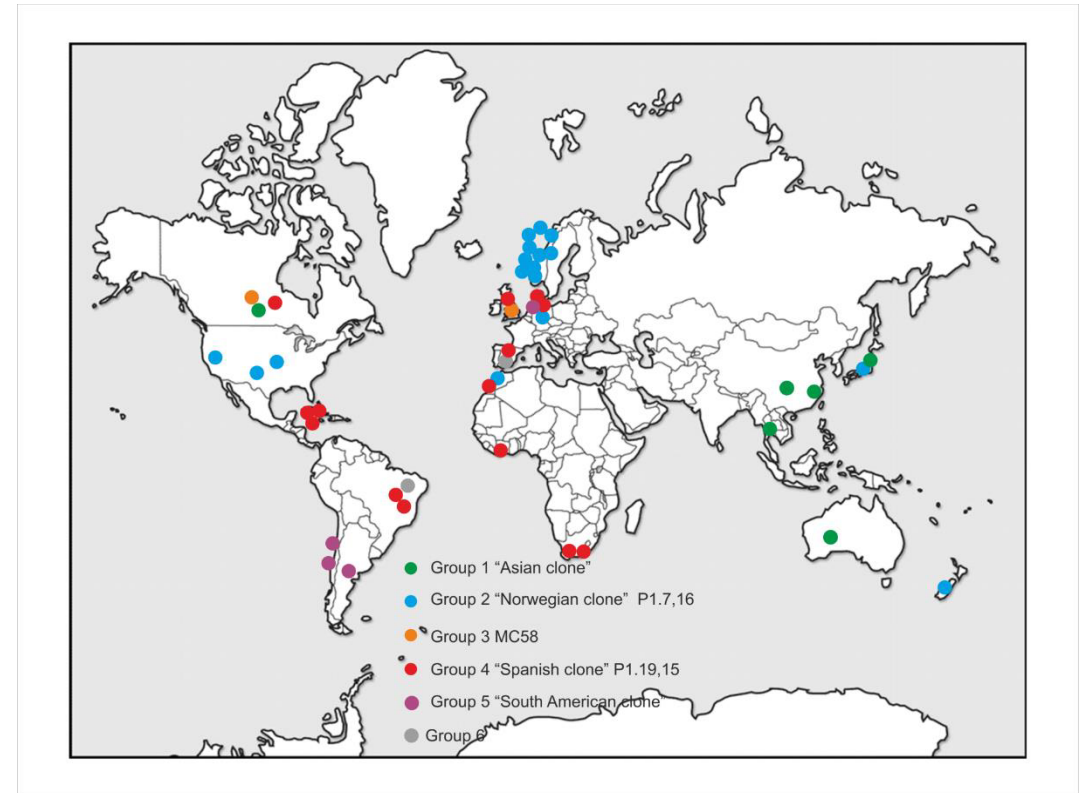
Bratcher, H. B., Corton, C., Jolley, K. A., Parkhill, J. & Maiden, M. C. (2014). A gene-by-gene population genomics platform: *de novo* assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes. *BMC Genomics* **15**, 1138.

Introduction to PubMLST

Global epidemiology of meningococcal disease

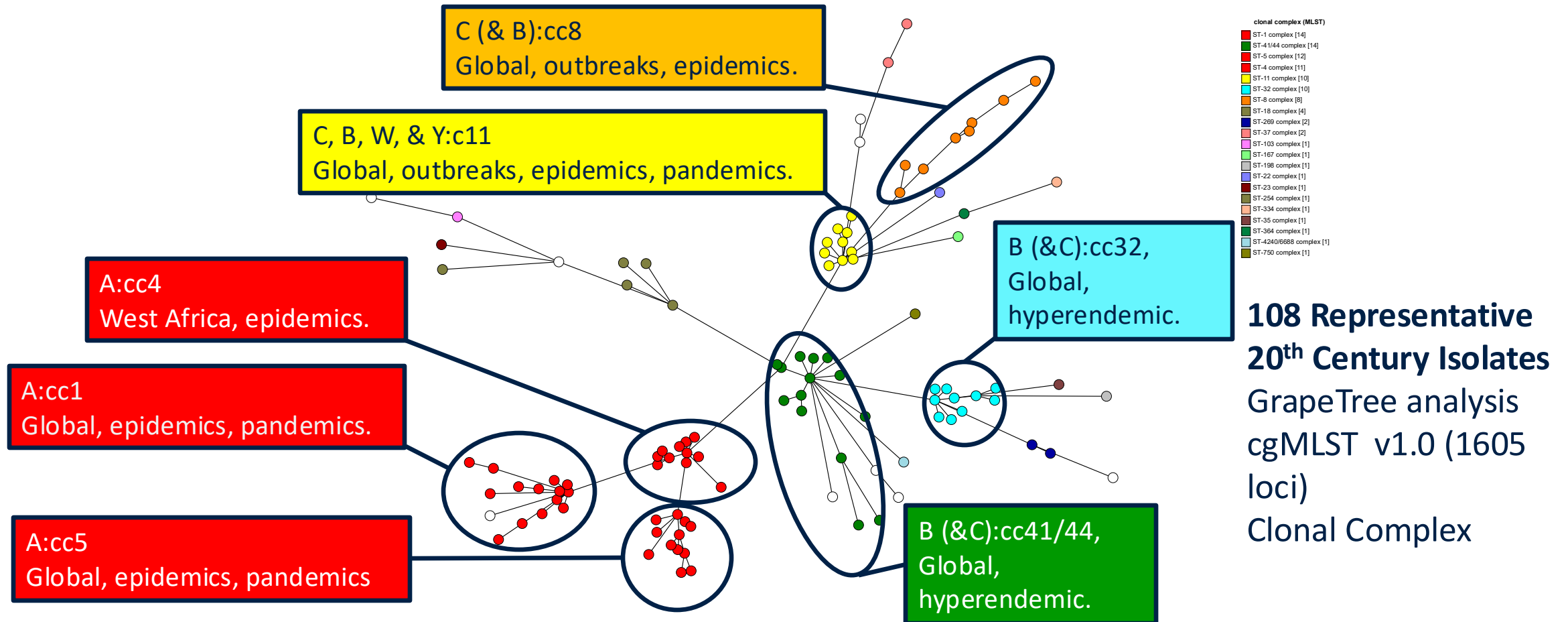
- **Endemic disease (multiple types):**
 - globally distributed;
 - 0.1 - 5 cases per 100,000.
- **Hyperendemic disease (e.g. B:cc32; B:cc41/44):**
 - Local (community), national, global;
 - ~10 cases per 100,000.
- **Disease outbreaks (e.g. C:cc11; W:cc11):**
 - local & institutional (army camp, university).
- **Epidemic/pandemic disease (e.g. A:cc5):**
 - Regional, national, & global;
 - 500 cases per 100,000, or more.

Harrison, L. H., Trotter, C. L. & Ramsay, M. E. (2009). Global epidemiology of meningococcal disease. *Vaccine* **27**, B51-B63.



Harrison, O. B., Bray, J. E., Maiden, M. C. & Caugant, D. A. (2015). Genomic Analysis of the Evolution and Global Spread of Hyper-invasive Meningococcal Lineage 5. *EBioMedicine* **2**, 234-243.

Age of Molecular Epidemiology: meningococcal disease in the 20th Century



Bratcher, H. B., Corton, C., Jolley, K. A., Parkhill, J. & Maiden, M. C. (2014). A gene-by-gene population genomics platform: *de novo* assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes. *BMC Genomics*. **15**, 1138.

Exercise 1: Diversity of meningococci in the 20th century (1)

- Go to <https://PubMLST.org> and log in.
- Click on **ORGANISMS**, then click on *Neisseria* spp.
- Click on **Isolate collection** then click on Search database
- In the right-hand top corner click on **Modify form**
- Click on the Filters button.
- On the main page in the **Publication:** box select:
 - Maiden et al. 1998 Proc Natl Acad Sci U S A 95:3140-5
- Click on **SEARCH**

Search or browse database



Enter search criteria or leave blank to browse all records. Modify form parameters to filter or enter a list of values.

Isolate provenance/primary metadata fields

id = Enter value... +

Filters

Publication: 1 of 306 selected

Project: Select options

× Epidemiological year:

× MLST profiles:

× Clonal complex (MLST):

Private records:

☐ Include old record versions

Add filter: Add

Display/sort options

Order by: id ascending

Display: 25 records per page

Action

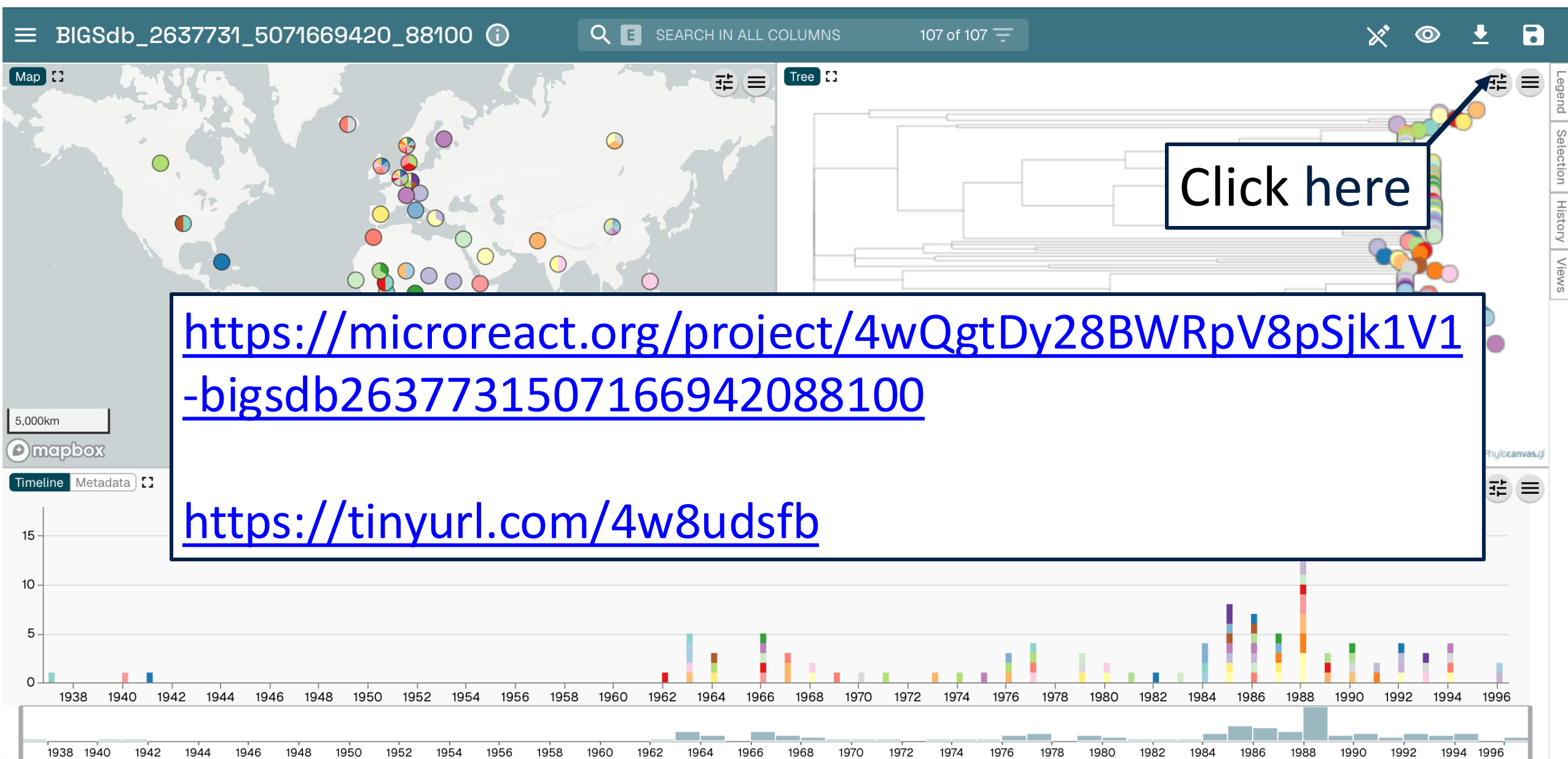
RESET

SEARCH



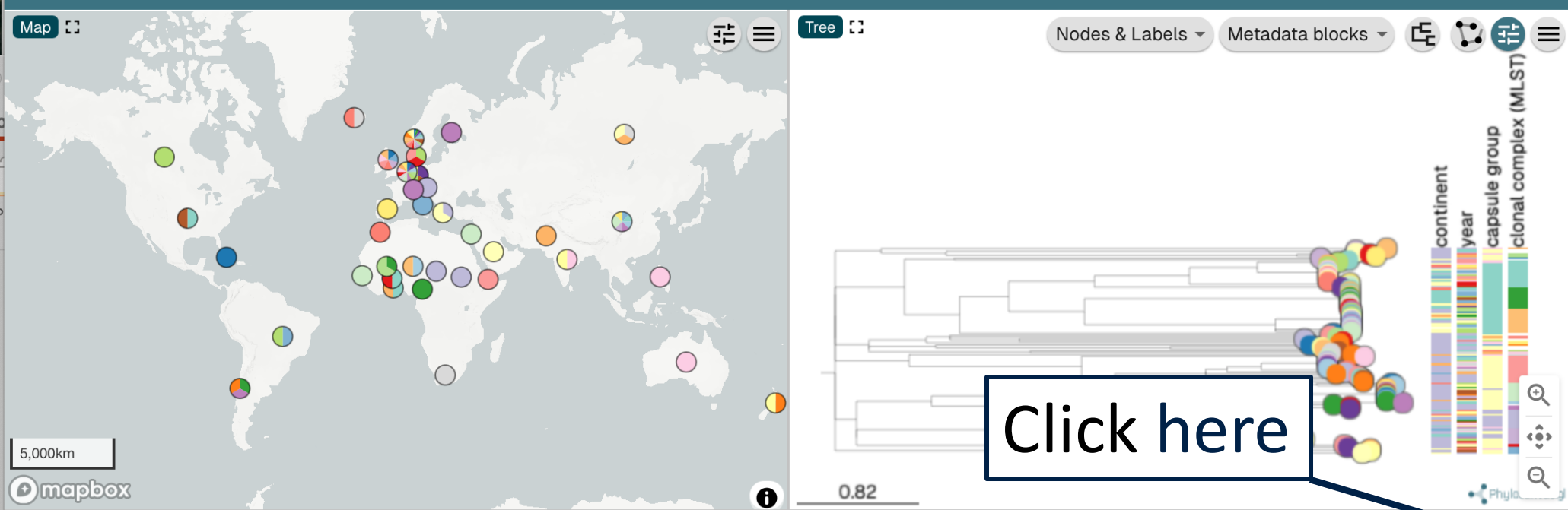
Exercise 1: Diversity of meningococci in the 20th century (2)

- Scroll to the bottom of the screen and click on **Microreact**
- Under Recommended Schemes select:
 - N. meningitidis cgMLST v3
- Under **Tree generation** select **profiles**
- Under Include fields choose:
 - continent, capsule group, ST (MLST), and clonal complex (MLST)
- Click on **SUBMIT**
- When the job has finished click on launch **Microreact**



Exercise 1: Diversity of meningococci in the 20th century (2)

- Under Metadata blocks select:
 - continent
 - year
 - capsule group
 - clonal complex (MLST)
- Click on Legend (top right hand corner)
- Scroll down, click on Africa, which years, capsule group, and clonal complexes are highlighted?



Legend

Colours by isolate

10

11-004

14/1455

20

26

38VI

50/94

71/94

88/03415

90/18311

91/40

94/155

106

120M

129E

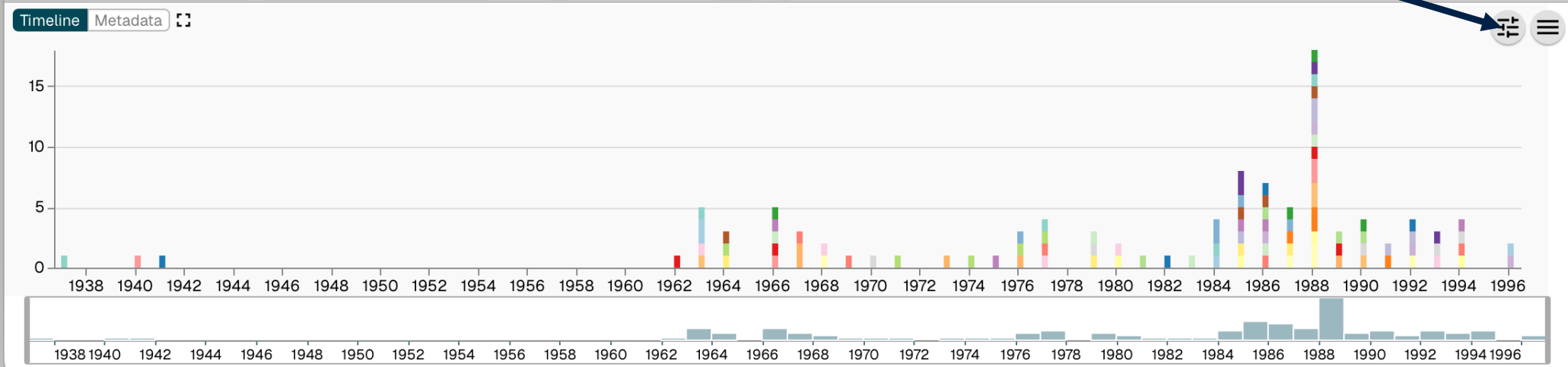
139M

153

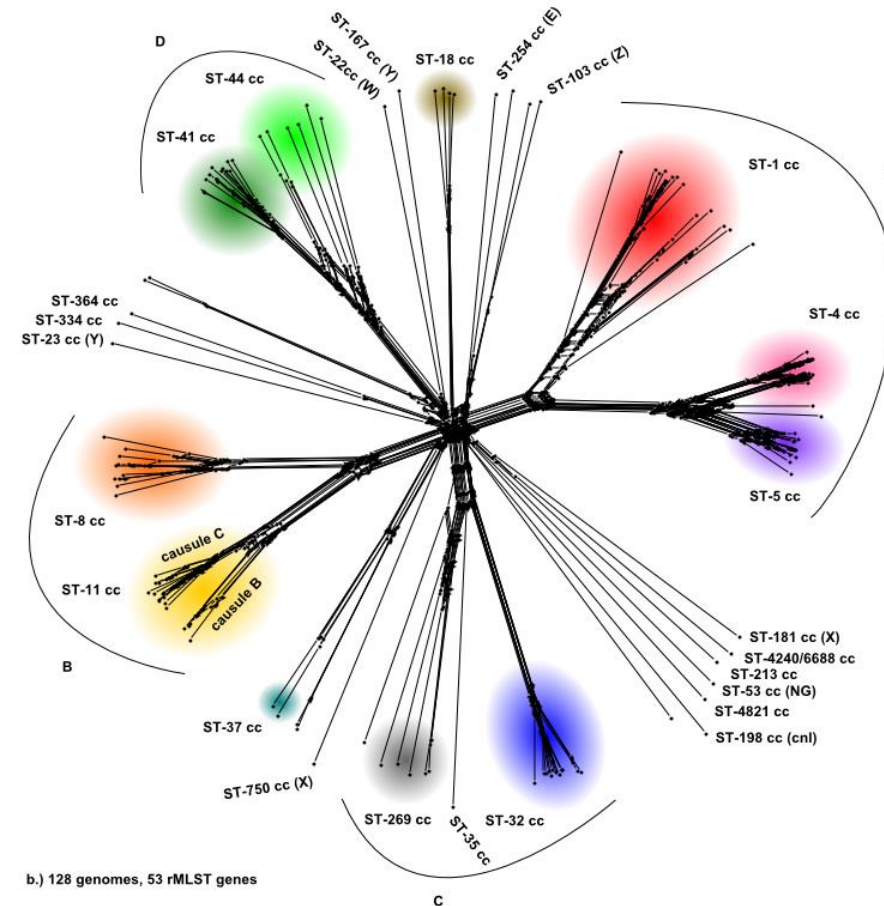
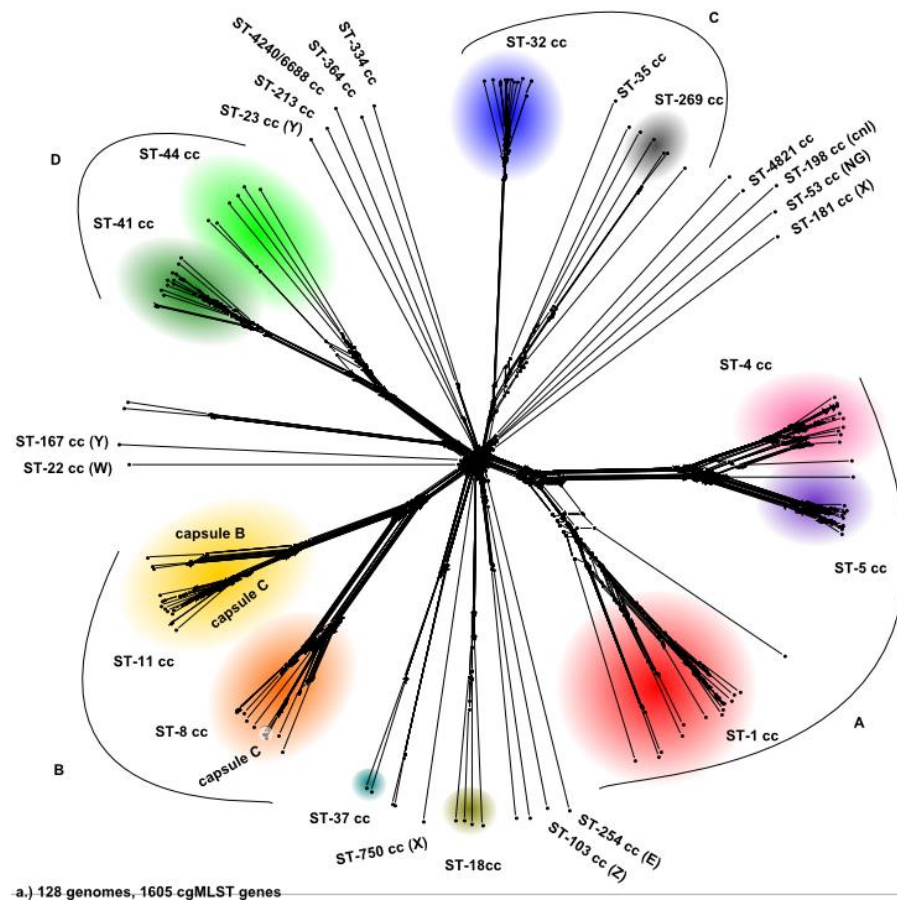
Selection

History

Views



Increasing data increases resolution



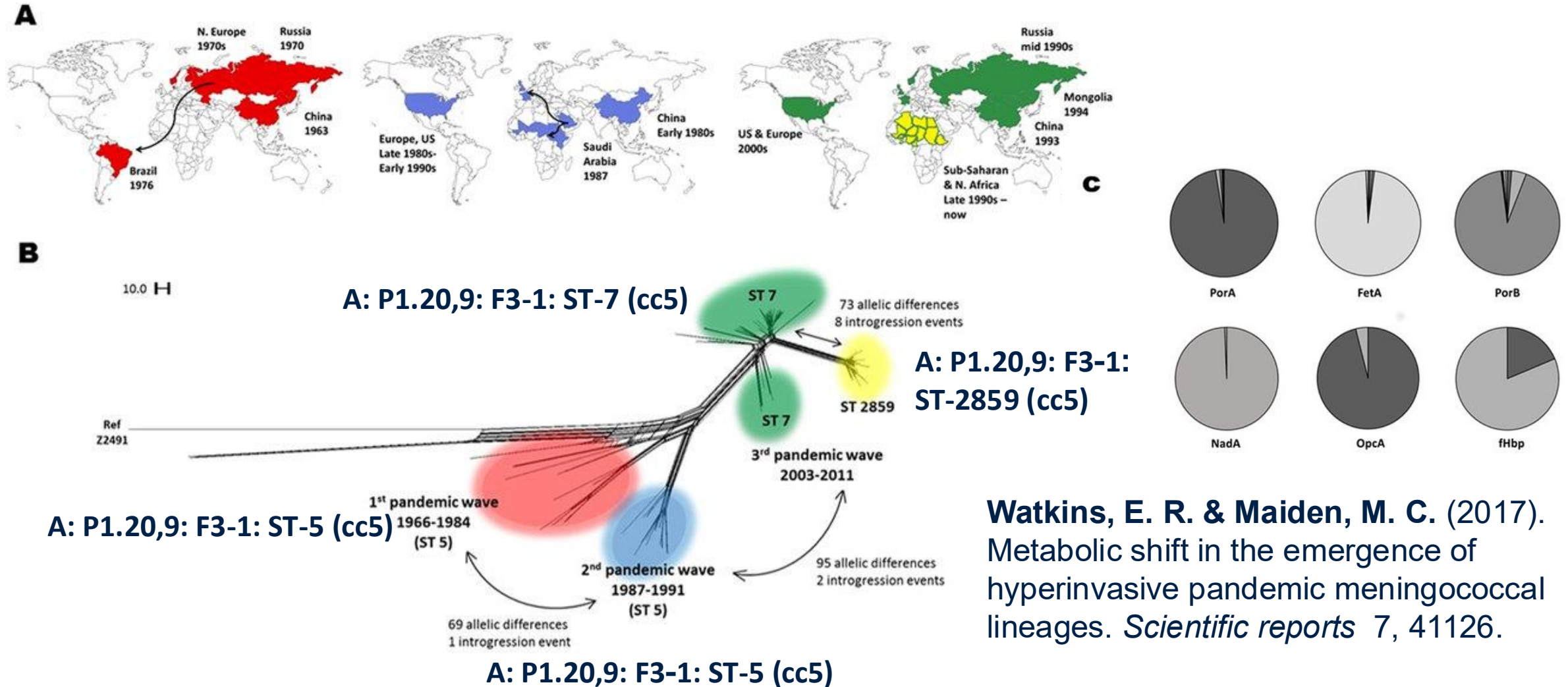
Bratcher, H. B., Corton, C., Jolley, K. A., Parkhill, J. & Maiden, M. C. (2014). A gene-by-gene population genomics platform: *de novo* assembly, annotation and genealogical analysis of 108 representative *Neisseria meningitidis* genomes. *BMC Genomics* **15**, 1138.

Increasing resolution

The following take you to the same analysis with

- rMLST:
 - <https://microreact.org/project/sM8Z6SSzgYGnNKDbZn7jTS-bigsdb2586417359492399828260>
 - <https://tinyurl.com/y7eyc7tk>
- cgMLST v 3.0:
 - <https://microreact.org/project/o9uiPWygnHqgjQQAA4KJLk-bigsdb2659350099187952922299>
 - <https://tinyurl.com/ymwdmwm8>

A:cc5 Pandemics stability over time



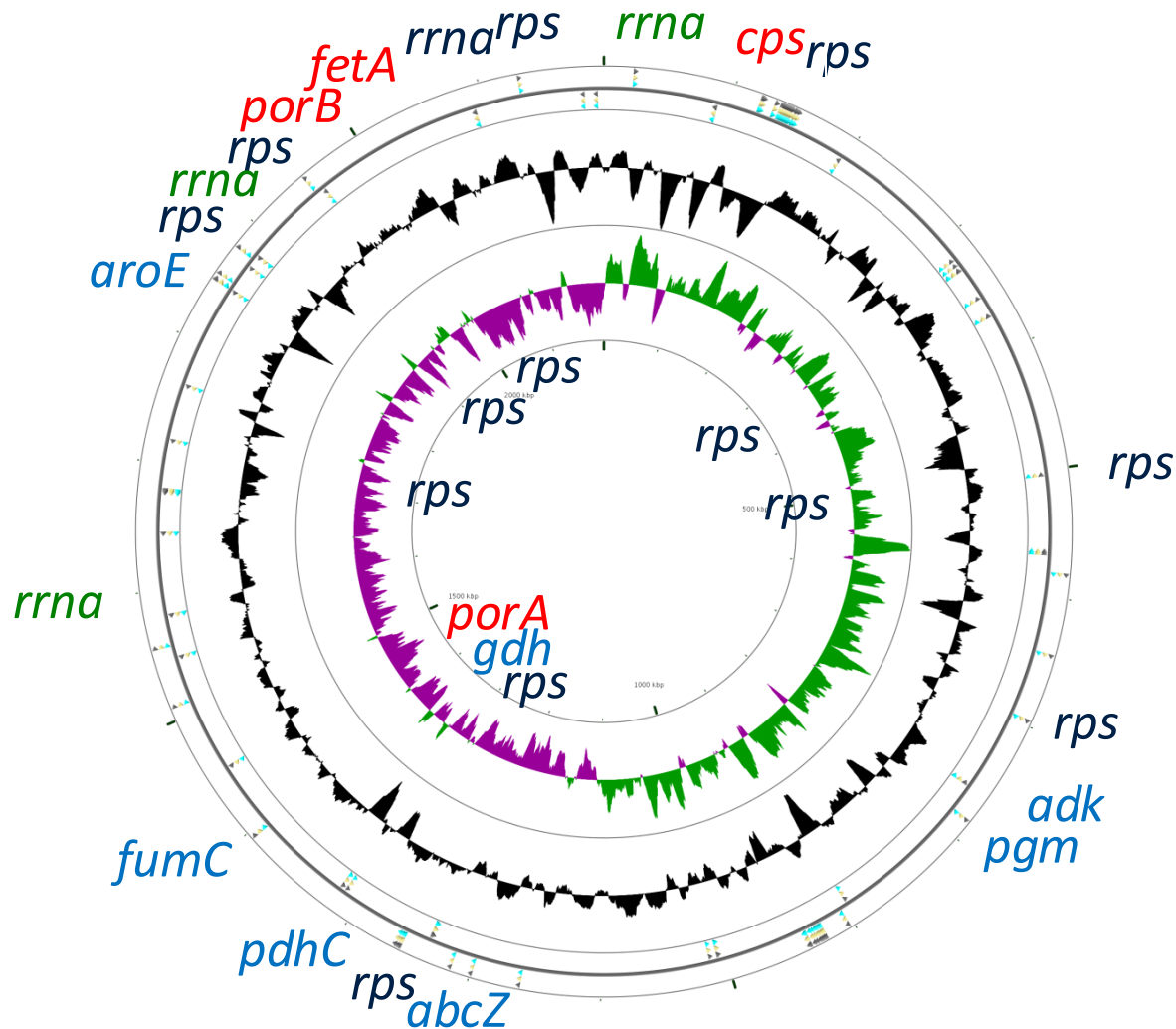
The evolution and spread of the A:cc5

A Microreact analysis of these isolates:

- <https://microreact.org/project/1kh6ruNZxaiobbZszMMKYH-bigsdb3991163873630764724981>
- <https://tinyurl.com/phfc5mj9>

Use this analysis to explore the evolution and spread of S:cc5 meningococci.

Summary: typing targets in the meningococcal genome



Antigen genes:

- Capsule (*cps*), serogroup;
- OMPs (*porA*, *porB*, *fetA*), serotype, subtype fine type;
- BAST antigens.

Housekeeping genes:

- Ribosomal RNA genes (*rrna*) genus identification;
- Ribosome protein genes (*rps*), species and sub species characterisation (rMLST);
- MLST (metabolism), subspecies identification
- cgMLST, high-resolution typing.

Antibiotic Resistance genes.

Session 8: Translation and application: 'Germs, Genes, and Genomes', exemplified by the control of meningococcal disease in Africa

Molecular Approaches to Clinical Microbiology in Africa 2025

Martin Maiden

Department of Biology



UNIVERSITY OF
OXFORD

Meningococcal disease: still with us

Peltola, H. (1983). *Rev Infect Dis* 5, 71-91.



Vieusseux, G. (1806). Mémoire sur la maladie qui a régné à Genève au printemps de 1805. *J Med Chir Pharm* 11, 163-182.

Nigeria records 961 meningitis cases, 56 deaths

5th October 2022



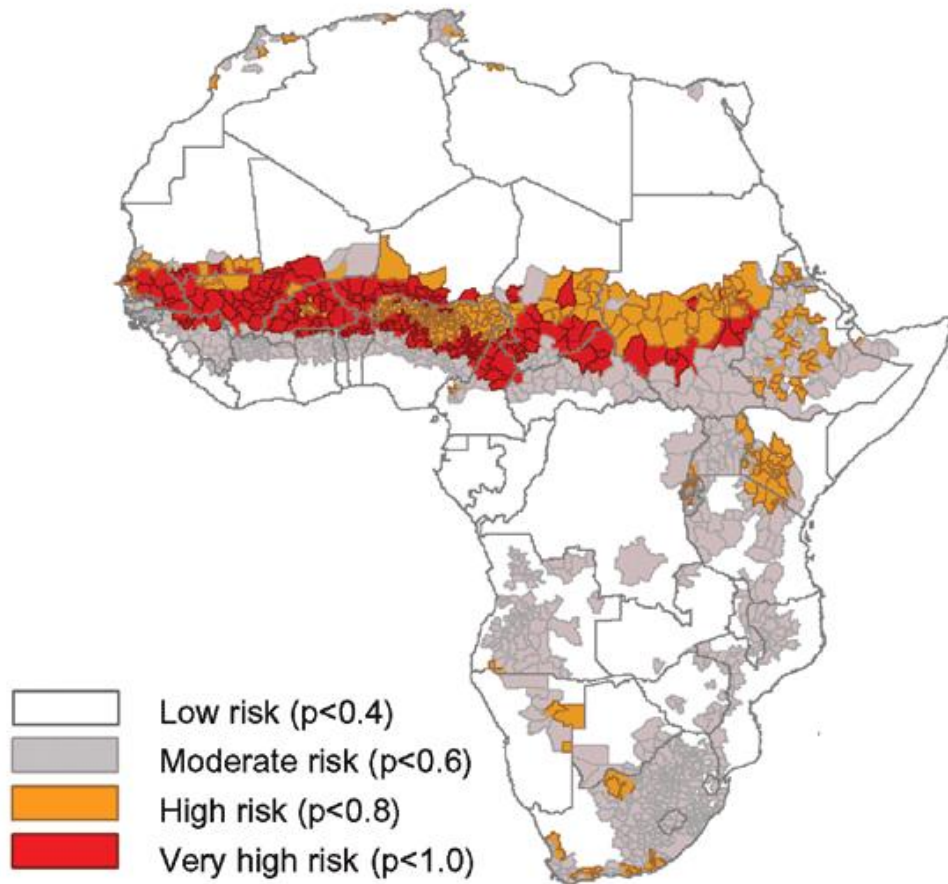
NCDC Director-General, Ifedayo Adetifa

By Lara Adejoro

<https://punchng.com/nigeria-records-961-meningitis-cases-56-deaths/>

News Item 5th October 2022; accessed 27th January 2023.

The meningitis belt



Meningitis belt extends from Ethiopia to Senegal.

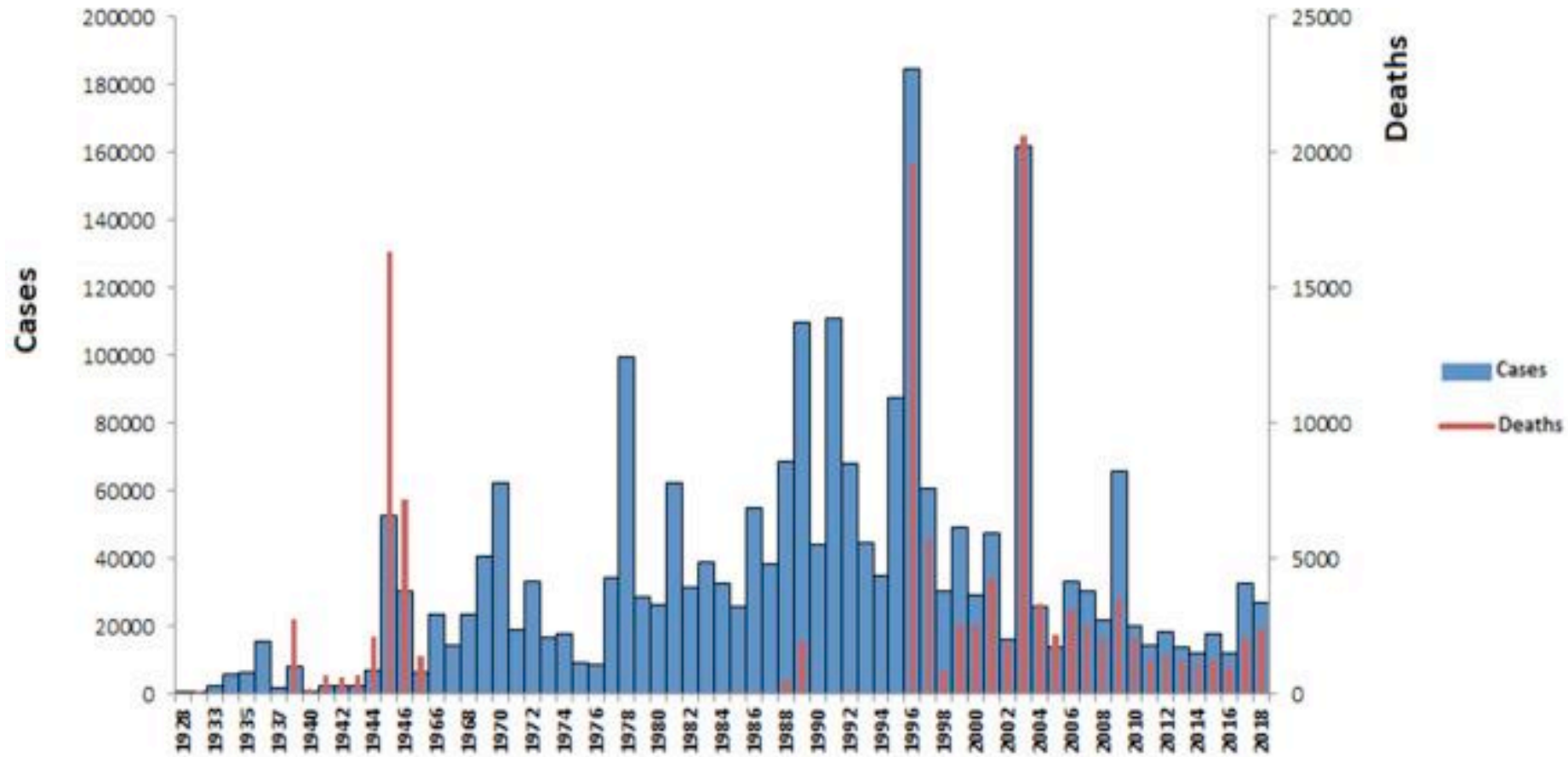
Sudan, Ethiopia, Chad, Niger, Northern Nigeria, Burkina Faso, and Mali are considered hyper-endemic.

- **1905**, first documented epidemic, Northern Nigeria.
- **1919-1924**, second cycle, >45,000 deaths, Northern Nigeria.
- **1935-1937**, third cycle, Nigeria, 6,456 deaths.
- 1951-60, 340,000 cases, 53,000 deaths.
- 1996-1997, 300,000 cases, 30,000 deaths

Molesworth, A. M., Cuevas, L. E., Connor, S. J., Morse, A. P. & Thomson, M. C. (2003). Environmental risk and meningitis epidemics in Africa. *Emerg Infect Dis* **9**, 1287-1293.

Greenwood, B. (1999). Manson Lecture. Meningococcal meningitis in Africa. *Trans R Soc Trop Med Hyg* **93**, 341-353

Meningococcal disease in Africa



Mazamay, S., Guegan, J. F., Diallo, N., Bompangue, D., Bokabo, E., Muyembe, J. J., Taty, N., Vita, T. P. & Broutin, H. (2021). An overview of bacterial meningitis epidemics in Africa from 1928 to 2018 with a focus on epidemics "outside-the-belt". *BMC Infect Dis.* **21**, 1027.

MenAfriVac®: a solution to epidemic meningococcal disease in Africa

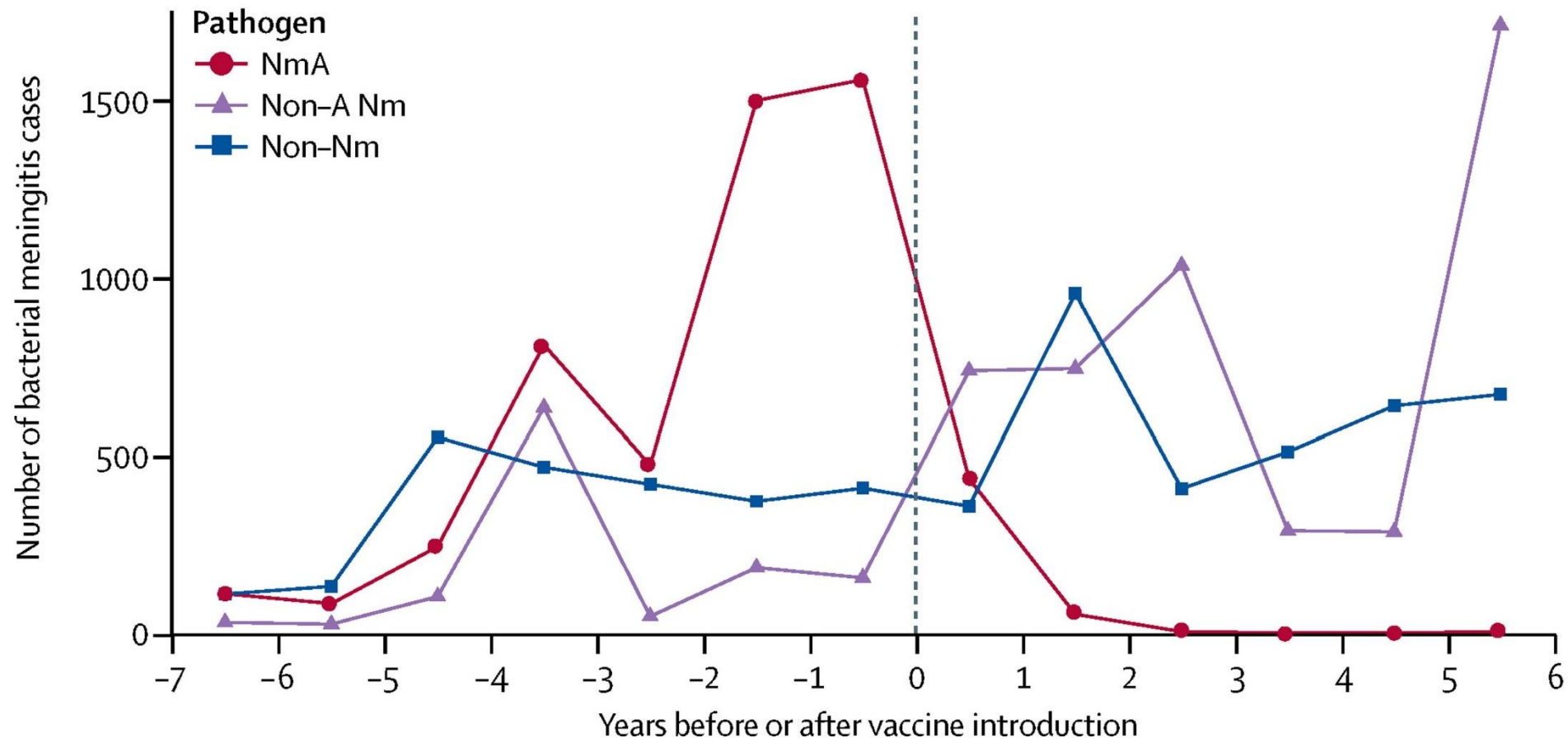
Meningitis Vaccine Project (MVP):

- Gates funded, partnership with WHO and PATH;
- Innovative development – Northern technology, southern manufacturer;
- Low cost (less than \$1 per dose, for sustainable use);
- MenAfriVac, a polysaccharide-protein conjugate vaccine, introduced 2010 in Burkina Faso;
- **Everyone aged 1-29 years immunised;**
- **Immediate reduction in disease levels.**

Diomande, F. V. K., Djingarey, M. H., Daugla, D. M., Novak, R. T., Kristiansen, P. A., Collard, J. M., Gamougam, K., Kandolo, D., Mbakuliyemo, N., Mayer, L., Stuart, J., Clark, T., Tevi-Benissan, C., Perea, W. A., Preziosi, M. P., LaForce, F. M., Caugant, D., Messonier, N., Walker, O. & Greenwood, B. (2015). Public Health Impact After the Introduction of PsA-TT: The First 4 Years. *Clinical Infectious Diseases* 61, S467-S472.

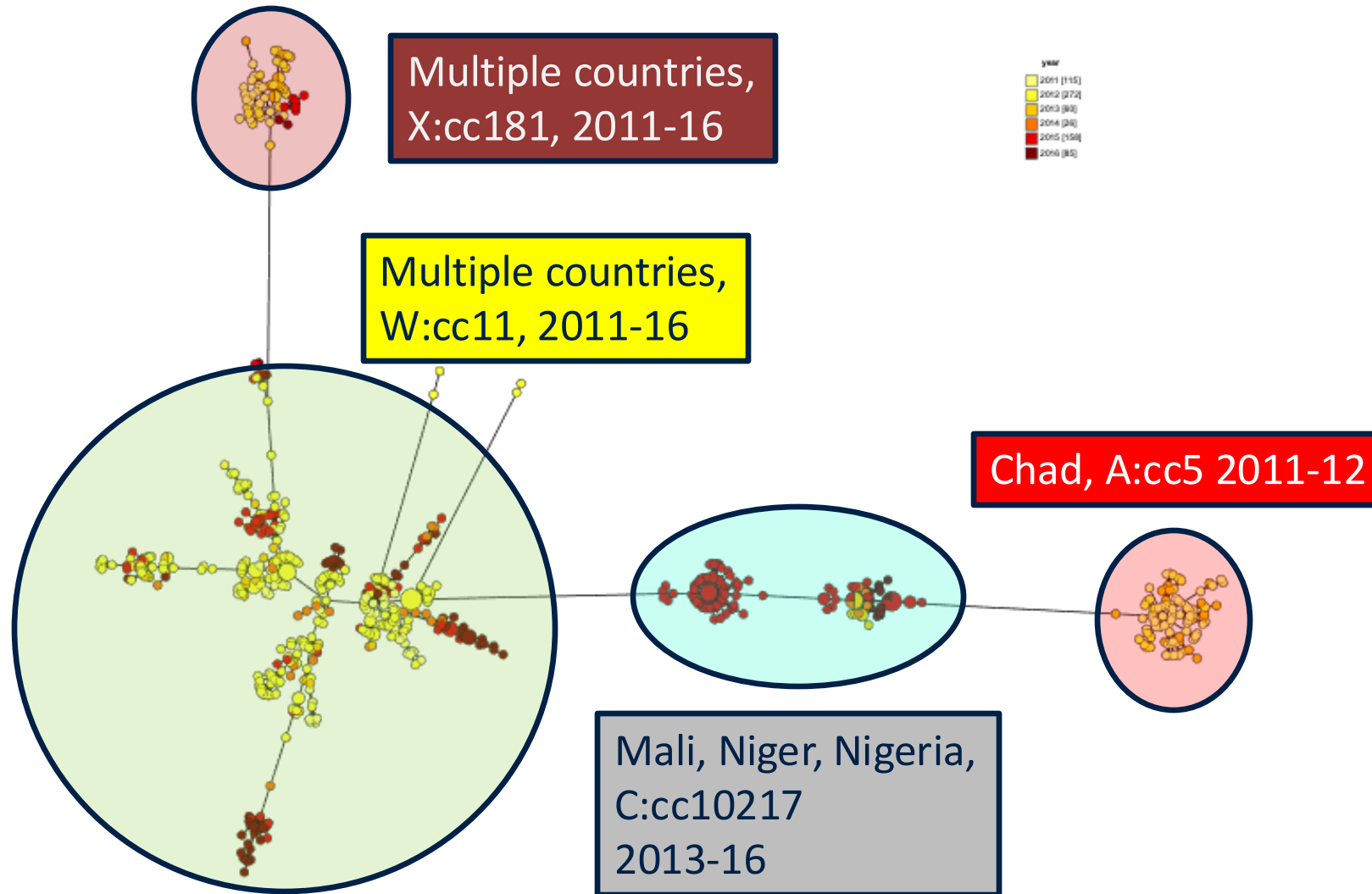


MenAfriVac®: impact on confirmed meningitis cases



Trotter, C. L., Lingani, C., Fernandez, K., Cooper, L. V., Bitá, A., Tevi-Benissan, C., Ronveaux, O., Preziosi, M. P. & Stuart, J. M. (2017). Impact of MenAfriVac in nine countries of the African meningitis belt, 2010-15: an analysis of surveillance data. *Lancet Infectious Diseases* **17**, 867-872.

Genomes: African IMD Isolates 2011-2016



Data from:

Topaz, N., Caugant, D. A.,
Taha, M. K., Brynildsrud, O.
B., Debech, N., Hong, E.,
Deghmane, A. E., Ouedraogo,
R., Ousmane, S.,
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Lafourcade, B. M., Diarra, S.,
Fox, L. M. & Wang, X. (2019).
Phylogenetic relationships
and regional spread of
meningococcal strains in the
meningitis belt, 2011-2016.
EBioMedicine **41**, 488-496.

Analysis:

PubMLST, GrapeTree, cgMLST
1.0

Concluding remarks

- Global surveillance of infectious disease agents is possible and being done, however, there remain challenges:
 - Timely data collection and reporting;
 - Data quality, both sequences and accompanying data;
 - Unified nomenclatures have to be established and maintained;
 - Analysis techniques have to be suitable for the organism in question;
 - Data sharing and reuse have to be both effective and appropriate.